

Creating Career Women: Access to Teaching Jobs and Women's Long-Run Outcomes

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October 1, 2026

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Abstract

How do career opportunities change the trajectory of women's work and family lives? I study this question in the context of teaching in the early twentieth century, which accounted for 25% of women's white-collar work in 1920. I digitize records from Illinois teacher-certification exams (1916–1923) and link these to the full-count censuses (1910–1940) to implement a regression discontinuity design around the exam's passing threshold. I find that marginal passers are 13.6 percentage points (36%) more likely to ever be teachers and 9.6 percentage points (16%) more likely to ever be in the labor force; for marginal failers, the alternative to teaching was typically not a different occupation, but to not enter the labor force at all. Marginal passers continued to be more likely to remain in the labor force for the next 20 years. Long-run persistence in the labor force was concentrated among those who remained in teaching and in part among women who remained in the labor force after marriage when institutional features permitted. Those who remained as teachers also acquired additional schooling. Access to teaching also delayed and reduced marriage and fertility, but did not change the characteristics of whom women married. These results provide new quasi-experimental evidence that career opportunities in early adulthood can induce women into sustained labor force participation, with corresponding implications for their family outcomes.

1 Introduction

Women’s work and family lives, and how they interacted, changed dramatically over the course of the twentieth century in the US. Female labor force participation tripled between 1900 and 1990; 60% of this growth was among married women in white-collar and professional work (Figure 1). This growth reflected a fundamental shift in how women engaged in the labor market, both in their persistence in the labor force after marriage and in the types of work they did, which increasingly rewarded education and experience (Goldin, 1990). Occurring alongside these changes was, with the exception of the baby-boom era, women’s significant decline in fertility.

Work histories suggest that new entrants to the labor force, and the kinds of occupations they first entered, were key determinants of this growth in labor force participation (Goldin, 1989). However, because women who pursue professional work differ in ability, motivations, and preferences, it remains unclear whether access to such work causally generated long-run persistence or instead drew in women who would have worked anyway—and how it affected family outcomes. Existing papers address this by exploiting changes across places and over time in labor demand (Acemoglu et al., 2004; Fernández et al., 2004; Goldin and Olivetti, 2013), technology (Rashid, 2025; Feigenbaum and Gross, 2024; Greenwood et al., 2005; Vi-dart, 2024), reproductive technology (Goldin and Katz, 2002), and educational opportunities (Goldin, 2006; Doxey et al., 2026; Bald, 2025) to study women’s entry into work or specific occupations. These approaches capture changing opportunities together with the broader labor-market and institutional changes that accompany them.

In this paper, I use newly digitized historical records on teacher certification exams to exploit quasi-random, individual-level variation in women’s access to teaching in the early twentieth century—holding fixed the labor demand, technologies, and institutions that shaped women’s opportunities in aggregate—to estimate how access to professional work affected women’s labor-market trajectories and subsequent education, marriage, and fertility outcomes. Access to a teaching career provides a valuable case study because it was one of the few socially acceptable and highly sought-after professional occupations available to women. In 1920, one-fourth of women’s white-collar work was concentrated in teaching, making it the single largest white-collar occupation for women (Figure 2). For many women, a barrier to teaching was a barrier to professional work itself. Teaching was also distinct among occupations available to women in that it had several career-like attributes, including credentialing requirements, relatively high pay that scaled with experience, pensions, and required ongoing professional development.

Teaching is also one of the few occupations in which mandated entry barriers generated

quasi-random variation in who gained access. In 1920, all states except Vermont offered an examination pathway to attain teaching credentials (Cook, 1921).¹ These systems imposed strict passing criteria that generated individual-level variation in access to teaching. I focus on the certification system in Illinois, where applicants completed thirteen examinations and were statutorily eligible for teaching positions if they obtained a cumulative score of 975 points, with no subject score below 60 percent. Using a regression-discontinuity design, I compare the life-cycle outcomes of women whose first-attempt cumulative scores fell just above and below the 975-point threshold.

To implement this design, I scan and digitize 24 volumes of elementary school teacher-certification examination ledgers spanning 1916 to 1923. The records cover all Illinois counties except Cook County² and contain approximately 48,000 applicants with detailed information on each applicant and her subject scores. During these years, about half of new elementary certificates issued to women outside Cook County came through examinations (Figure 3). I focus on the 79 percent of applicants who were women. Because applicants were young—18.2 years old on average, with fewer than one percent married—I link them to the census immediately preceding the examination, when they are generally observed in their childhood households, and then follow them forward through the 1920, 1930, and 1940 censuses to observe their long-run labor market, education, marriage, and fertility outcomes.

The main threats to my regression discontinuity design are endogenous retaking and manipulation around the passing threshold, but the structure of the examination and grading process allows me to address both. First, applicants could retake the examination, and the decision to retake is endogenous and can bias estimates (Jepsen et al., 2016). Because applicants could also retain passing subject scores and retake only the subjects they had failed, subsequent attempts differed in nature from the first. I therefore focus on first-attempt scores. Second, I find evidence that graders increased candidates’ minimum subject scores to the required 60 percent when this change would allow them to pass the exam. Because graders disproportionately increased minimum scores for candidates whom the increase would put in a passing position, this behavior creates bunching in both the minimum-score and cumulative-score distributions. Unlike in most settings with manipulation, however, this pattern allows me to detect the relevant manipulation and identify the affected test takers. I exclude applicants whose minimum subject score equals 60, after which the distribution of cumulative scores is smooth around the threshold. Other concerns about strategic manipulation are mitigated by the institutional setting, in which examinations were graded

¹States varied in whether examinations were required in addition to or in place of other educational credentials.

²Cook County was exempt from the statewide certification system.

anonymously by the State Board of Examiners, which limited graders' ability to influence scores for specific named applicants.

Passing the teacher examination substantially increased women's entry into both teaching and the labor force. Marginal passers are 13.6 percentage points (36%) more likely ever to be observed as teachers in a subsequent census and 9.6 percentage points (16%) more likely ever to be observed in the labor force. I cannot reject that these coefficients are equal, consistent with women on the margin of entering teaching being on the margin of entering the labor force at all. These are intent-to-treat estimates. Because marginal failers could retake the examination, passing on the first attempt raises the probability of ever clearing the cumulative threshold by about 50 percentage points, so the implied effects of certification itself are roughly twice as large.

These effects persisted for more than two decades. By 1930, marginal passers were 8.0 percentage points more likely to be teachers and 7.3 percentage points more likely to be in the labor force. Even by 1940, 17 to 24 years after the examination, they remained 8.5 percentage points more likely to be teachers and 8.9 percentage points more likely to be in the labor force. Pooling the 1930 and 1940 censuses, women are 44% more likely to be teachers, and 19% more likely to be in the labor force in the long run. There is also suggestive evidence that women had higher wage income (\$76.3, or 28%) by 1940, which is explained entirely by their greater extensive-margin labor force participation. This is strong evidence that access to teaching generated persistent careers for women.

I next examine for whom and under what conditions the labor-force effects persisted. First, long-run employment was driven almost entirely by persistence in teaching as opposed to entry into other occupations, consistent with the similar magnitudes of the teaching and labor-force effects across census years. Second, roughly half of the long-run labor force participation effect (measured in 1940) was concentrated among women who continued to work after marriage. Teaching did not crowd out married women's work in other occupations. If anything, there is suggestive evidence of additional entry into other white-collar work. Finally, education compounded the effects of initial career access. Marginal passers were more likely to acquire additional education after the certification examination,³ with the effect concentrated at the two-year college margin and among women who remained teachers through 1930 and 1940.

Although my sample size is insufficiently powered to support extensive heterogeneity analysis, two suggestive patterns emerge. First, whether women could sustain these careers after marriage depended on local restrictions on married women's work; I build on data

³Data on education at the time of the examination allow me to distinguish education attained before and after the examination.

collected in [Yang and Brough \(2026\)](#) to identify Illinois counties that likely operated teacher marriage bars, which were school-board policies that restricted married women from working as teachers. Marginal passers persisted in the labor force after marriage only in counties without a marriage bar. In counties with a marriage bar, the married-teacher effect disappeared, and women did not substitute into other work. These results suggest that marriage bars prevented women from remaining at work, and that women’s desire for continued career after marriage may have been stronger than aggregate labor force participation rates suggest.

Second, I find that the labor-force and income effects are concentrated among farmers’ daughters, who were disproportionately rural and had the fewest alternative white-collar opportunities. For women from non-farmer households, access to teaching did not newly draw them into the labor force, but shifted which occupation they entered in the labor force. While suggestive, these patterns highlight the disparate impacts of local labor-market conditions and family-compatible institutions in shaping women’s career and family outcomes.

Access to teaching also changed women’s family formation by both delaying and deterring marriage. Marginal passers were 8.5 percentage points more likely never to have married by 1930 and 5.0 percentage points more likely never to have married by 1940, consistent with marriage delay. They were also 7.1 percentage points less likely to have any children by 1940, and delayed their first birth by approximately 0.8 years. I find some imprecise evidence that the number of children, conditional on having children, decreased. Despite these large changes in marriage and fertility, I find little evidence that access to teaching changed the characteristics of whom women married. Husbands did not have higher income, education, or income rank relative to the woman’s father. Access to a professional career therefore changed whether and when women formed families, but did not generate gains in the marriage market; instead, the economic gains were concentrated among women themselves.

I connect my estimates to aggregate changes in LFP and in fertility across birth cohorts to show that expanded access to professional work can account for a meaningful share of these broad demographic shifts. Under a very conservative framework, I show that additional entry into the labor force, driven by marginal exam passers (those within 3 percentage points of the passing threshold), explains roughly 1 percent of the rise in prime-age labor force participation and 0.2 percent of the decline in completed fertility in downstate Illinois. Extending my estimates to other geographies and entrants in non-teaching white-collar work, I find that additional early-career entry accounts for 16 to 36 percent of the rise in labor force participation and 4 to 8 percent of the decline in fertility.

My results contribute to a growing literature studying how women’s economic opportunities affected labor market and family outcomes during the early-to-mid twentieth century. Seminal work exploits variation in female labor demand generated by World War II mobi-

lization to study the effects of women’s employment on wages (Acemoglu et al., 2004), labor supply (Goldin and Olivetti, 2013), fertility (Doepke et al., 2015), and the social preferences of the next generation (Fernández et al., 2004).⁴ More closely related to my work are a recent set of papers using the linked census microdata to study women’s entry to specific occupations. These include how workplace technologies expanded opportunities in clerical work (Rashid, 2025), displaced female telephone operators (Feigenbaum and Gross, 2024), and how proximity to nursing schools facilitated women’s entry into the profession (Bald, 2025). These papers exploit temporal and geographic variation in economic opportunities available to women, which by design combines individual effects with peer and network effects.⁵ In contrast, I identify a shock that determined women’s access to an occupation at the individual level, holding fixed labor markets and education institutions, as well as the norms and preferences that evolve with them.

Relative to existing work, my setting shows that career women are not pre-determined and can be induced by early career opportunities into sustained labor force participation lasting more than two decades. This is unlike settings in which women left the labor force after an occupational opportunity disappeared (Feigenbaum and Gross, 2024; Li, 2025; Rose, 2018) or upon marriage (Rashid, 2025). It also differs from Bald (2025), who finds that expanded access to nursing created persistent careers but primarily diverted women from other occupations. Instead, access to teaching drew marginal women into the labor force in early adulthood and kept them there in the long run.

In this way, my work speaks not only to the literature on women’s entry into work, but to the forces and institutions that sustained (or limited) women’s labor force participation over the life course, in particular their persistence after marriage. Prior work emphasizes changes that made married women’s employment more acceptable or more compatible with family life, including exposure to working women (Fernández et al., 2004), improvements in maternal health and infant feeding (Albanesi and Olivetti, 2016), and marriage bars (Goldin, 1988; Kim and Tsao, 2025; Yang and Brough, 2026). I show that labor force persistence was driven in large part by married women; however, this effect is observed only where marriage bars did not restrict their work. These findings suggest that demand for sustained work

⁴Recent work has questioned whether World War II mobilization led to women’s increased participation in the labor force by 1950. Using more granular data on the geography of war work, Rose (2018) shows that locations with the largest growth in war work did not exhibit sustained increases in female labor force participation by 1950. In a separate exercise, Briskin (2026) revisits Acemoglu et al. (2004) and Goldin and Olivetti (2013) using the full-count census and shows that many of the results, originally derived from the 1% census sample, are not robust to using the full-count data.

⁵We might think this distinction is particularly relevant if exposure to working women itself changes perceptions or norms around women’s work (Fernández et al., 2004; Fernández, 2013), or for market-mediated outcomes such as marriage.

in occupations with career-like qualities among married women was stronger than observed participation rates imply due to institutional restrictions.⁶

I also contribute to research on how women’s economic opportunities shape family formation. Existing work has considered the effects of contraception (Goldin and Katz, 2000, 2002; Bailey, 2006), education (Black et al., 2008; McCrary and Royer, 2011; Geruso and Royer, 2018; Doxey et al., 2026), and labor-market changes (Vidart, 2026; Rashid, 2025) on marriage, fertility, and the timing of childbearing. A related literature studies how women’s economic opportunities affect whom they marry (Shenhav, 2021; Rashid, 2025). I show that access to teaching substantially delayed family formation and reduced fertility, but produced no corresponding improvement in husbands’ education, occupational standing, or earnings. Instead, the gains from professional access accrued primarily through women’s own employment.

Finally, I contribute to the literature studying teacher labor markets and certification systems. This work often centers on how women’s outside options (Corcoran et al., 2004; Bacolod, 2007) or state certification systems (Angrist and Guryan, 2008; Orellana and Winters, 2025) shape teacher quality and supply. Two notable exceptions study teachers’ own outcomes using individual-level certification-exam thresholds. Deneault et al. (2026) find that narrowly failing a Texas exam delays entry into teaching at a cost of roughly \$10,000 over a decade, and Tsao (2025) uses Kentucky’s entry-exam cutoff to estimate the pay and working-conditions rents teachers earn over their next-best jobs. Relative to these papers, the alternative to teaching in my setting was often not to enter the labor force, highlighting the importance of teaching as a central female occupation in the early twentieth century.

The remainder of the paper is organized as follows. Section 2 provides institutional background. Sections 3 through 4 describe the data, record linkage, and sample. Section 5 presents the empirical strategy and validity tests. Section 6 presents the labor effects, Section 7 the family effects, Section 8 heterogeneity, and Section 9 robustness. Section 10 considers their aggregate implications. Section 11 concludes.

2 Institutional Background

2.1 Teaching and Women’s Work

Teaching was among the first occupations to open to women on a large scale, and had become a female-dominated profession by 1860. It feminized as the spread of public common schools

⁶I also find suggestive evidence that the marriage bar induced different family-formation responses among affected women, providing new evidence on the non-labor market consequences of the marriage bar.

raised the demand for teachers while urbanization and industrialization drew men toward higher-paying work (Ogren, 2005). Reformers of the common-school movement encouraged hiring women, arguing that women had a natural predisposition to teach since it was in their nature to rear children (Ogren, 2005). Teaching was therefore distinct from other female work. It offered entry into a respectable white-collar profession at a time when nearly all other white-collar work was closed to women. For the next sixty years, teaching remained the dominant white-collar occupation for women.

By 1920, teaching accounted for roughly 25 percent of all women’s white-collar work and 10 percent of women’s total employment, making it the single largest white-collar occupation for women.⁷

Teaching also paid comparatively well by the standards of women’s employment in this era. Using the occupational income score—median earnings in each occupation as of 1950—teaching ranks highest among the principal occupations open to women (Figure 2). Teaching’s relative pay was more modest in the 1920s, although still reasonable. In 1922, the average downstate Illinois teacher earned \$1,046 over a school term of roughly eight months, or about \$131 per month in session. To benchmark this against contemporary alternatives, I digitize the pay scales in the 1922 classification of the Illinois State Civil Service Commission, which records monthly salaries alongside stated education requirements for every state job class (Illinois State Civil Service Commission, 1922). A teacher’s in-session pay matched or exceeded the starting pay of the clerical classes requiring a high school education—stenographers, department clerks, bookkeepers—though spread over twelve months it fell toward the middle of that range (Appendix Figure B1). Whether this other, potentially comparable white-collar work was available, however, was also determined by geography. For rural women, teaching made up 43% of white-collar work, relative to 13% in urban settings (Appendix Figure B2).

Aside from current pay, teaching underwent several changes between 1915 and 1940 that increasingly professionalized the occupation. These changes helped make teaching uniquely career-like relative to other occupations available to women. In 1915, teachers across Illinois outside Chicago gained access to a statewide pension system. School districts also increasingly adopted salary schedules that rewarded experience and additional training, while higher teaching credentials were typically associated with higher pay. Teachers were required in their first years, and then encouraged for more tenured teachers, to update their skills and qualifications through teacher institutes, summer coursework, and additional normal-school training. Together, these institutions created incentives to remain in teaching and invest in occupation-specific skills.

⁷Author’s own calculation using the 1920 full-count census.

At the same time, teaching (as well as other white-collar work) was considered incompatible with family life for many. Throughout this period, many school boards operated, either through formal or informal channels, the marriage bar, which limited married women’s ability to work upon marriage (Goldin, 1988). In Illinois, by 1942, around one-third of school districts responding to a survey reported marriage as potential grounds for dismissal. However, not all places operated the marriage bar, and how it operated depended on settings; for example, often school boards enforced the ‘hire-bar’ which prohibited the hiring of new married teachers more strictly than the ‘retention-bar’ in which teachers were dismissed upon marriage. Whether school systems operated marriage bars also depended on the local supply of teachers, which was often more constrained in rural areas.

2.2 The Illinois State Examination System

Illinois introduced its statewide examination system in 1914, replacing a fragmented system of county-operated exams (State of Illinois, 1915). The state system was part of a broader effort to improve teacher quality by centralizing and professionalizing teacher qualifications. Officials described the certificating law as the beginning of a “new and a better era” in Illinois education, and the 1914 law as a major step toward “keeping out the incompetent” and attracting competent teachers (Illinois Department of Public Instruction, 1919, 1923, p. 3).

The 1914 law created third-grade, second-grade, and first-grade elementary certificates.⁸ The second-grade certificate, which is the primary credential in my empirical setting, was valid for two years and renewable on evidence of successful teaching, professional training, and professional growth (State of Illinois, 1915, p. 8).⁹ The exam therefore created a regular, statewide route into teaching, but it was not meant to be a complete substitute for further training.

The examination system remained central between 1916-23 (Figure 3). About half of new elementary certificates issued to women outside Cook County came through examinations. Officials recognized that raising standards too quickly could restrict teacher supply, especially during and after World War I. Ultimately, in 1924, the board of examiners raised the passing threshold for the second-grade certificate from 75 to 80 percent, and in turn the number of

⁸Certificate “grades” denote tiers of the credential, not the school grade levels taught.

⁹The third grade certificate was a 1-year certificate available to individuals failing the second-grade certificate, renewable only once for an additional year; some people applied directly to the third grade certificate, but this accounted for less than 10% of applications. In 1919, this certificate was abolished, leaving those who failed the exam eligible for an ‘emergency certificate’ which was non-renewable and valid for 1 year. Neither the third grade (‘provisional certificate’) nor the ‘emergency certificate’ offered a career-like pathway to teaching due to its non-renewable nature.

new elementary teachers entering by exams fell precipitously.

2.3 Exam Grading and Administration

Examinations were held two to three times each year at each county seat. Applicants were assigned an identification number, which they wrote on their exam papers in place of their names, so that exams could be graded anonymously. Applicants had to pay \$1 (equivalent of \$19–\$30 today) to write. Written exams were mailed to the Illinois State Examining Board, which graded every exam centrally and recorded scores in a ledger that was then shared with counties.¹⁰ The formal examination rules specified sealed envelopes for each half-day's questions, separate desks for applicants, no communication among applicants, and central return of sealed manuscripts to the Examining Board (*Illinois State Examining Board, 1940*, pp. 7–9).

Candidates sat for exams in thirteen subjects (*State of Illinois, 1915*). Between 1916 and 1923, to pass a candidate had to obtain an average of at least 75 percent across all subjects (975 points). In principle, candidates were only eligible to receive a certificate if they had no subject grade below 60 percent.¹¹

Candidates could re-sit the examination at a later sitting, and retaking was common; roughly 40 percent of test-takers sat the exam more than once. Subject scores above 75 percent carried over from one attempt to the next, so returning candidate only needed to re-sit for the subjects she had not yet cleared. This carry-over makes scores on later attempts difficult to compare across candidates, since after the first attempt each retaker effectively faces a person-specific passing threshold that depends on which subjects she has already passed.

Passing the certification exam was not sufficient to receive credentials. Ledgers of exam scores (which include the names and scores of all county applicants) were mailed to County superintendents, who then had discretion in hiring teachers and could impose additional criteria if they saw fit. For example, teaching applicants had to additionally prove their good 'morals' as vouched for by 3 character references.

Therefore, because not all passers were offered (or accepted) teaching positions and some who failed found alternative routes into the profession, obtaining the average on the first-

¹⁰The board would enlist the help of educational professionals to assist in grading, but additional details on who specifically was recruited to grade exams are not known; though often it seems that these were education professionals in Springfield. How graders were assigned to exams is also unknown.

¹¹The 1915 statute specifies subjects and certificate requirements. The numerical passing rule appears in State Examining Board administrative materials and in the examination ledgers. Later administrative rules required a general average of 80 with a minimum of 50 for certificates earned by examination (*Illinois State Examining Board, 1940*, p. 8).

attempt write is best understood as a *fuzzy* treatment for access to teaching, and my regression discontinuity estimates capture intent-to-treat effects.

2.4 Alternative Pathways to Teaching Credentials

Normal schools and colleges could act as alternative credentials for students who either did not take, or did not pass, certification exams. Normal schools were distinct from colleges. They were teacher-training institutions, often with one- or two-year training programs, and some with more extensive 4-year BA options, that combined academic coursework with pedagogy and practice teaching.

In Illinois, students who attended two or more years of normal school or college could receive credit for this attendance and would be eligible for the second-grade teaching certificate, without taking the teacher exam. Completion of Normal School could also make candidates eligible for school positions, such as in the high school, specialist teachers, and principal, which would be routes not open to candidates gaining credentials through the exam alone.

Provisional and temporary certificates were another route into teaching, but they offered less security. Between 1916-1918, the third-grade certificate (or ‘provisional certificate’) existed and was available specifically for people who failed the second-grade certification again. It was valid for 1 year, eligible for renewal for an additional year, but could not be renewed a second time. In 1919, it was eliminated, leaving only the ‘emergency’ certificate, which was non-renewable, open for only 1 year, and open to those not writing the exam. Officials described these alternatives and temporary certificates as compromises needed to bolster teacher supply but ultimately phased out as standards continued to rise ([Illinois Department of Public Instruction, 1919, 1923](#), p. 3).

Nonetheless, these alternative pathways provide important context for understanding the RD design. While failing the exam did not make teaching impossible, it made entry more costly (entry through Normal school) or temporary (entry through emergency or, for some years, third-grade certificates).

3 Data

3.1 Examination Ledgers

My primary data source is examination ledgers obtained from the Illinois State Archives (Appendix Figure [B5](#) shows an example page). These ledgers record the results of the teacher certification examinations administered across all Illinois counties except Cook County, which

was exempt from the system. Each ledger corresponds to 1 exam sitting (unique at the month-year level). Each ledger entry records the candidate’s name, age, city of residence, years of education at the time of the examination, the date and county of the exam, and their exact score on each of the 13 subject exams.

Several information fields are typewritten, including names, ages, residence, and baseline education. The subject grades were handwritten. I digitized the ledgers using a large language model (Gemini 3.0-flash). Because each page is organized into grids with faint row and column boundaries, I first preprocess the page images to align and superimpose the column and row boundaries before passing them to the model; this step substantially improves transcription accuracy on the tabular score data.

3.2 Census Data

I link examination records to the full-count censuses of 1910, 1920, 1930, and 1940. First, the 1910 and 1920 censuses (depending on exam year) provide baseline household characteristics for candidates, including urban versus rural residence, the occupational score of the household head, the working status of the mother, and whether any woman in the household was in the labor force. The 1920 (for exams taken in 1916–1919), 1930, and 1940 censuses provide the relevant short- and long-run labor and family outcome variables. Below, I describe key variables in more detail.

- **Labor outcomes:** Teacher (occupation 93); other white-collar work occupation (occupations 1-99, excluding 93 and 200-490), other occupation, and labor force participation, and own income (reported only in 1940)
- **Marriage:** Marital status (specifically, never married)
- **Fertility:** Presence and Age of children in the household (used to estimate Age at first birth)
- **Spousal characteristics:** Husband’s education, occupational score, income (available in 1940)

3.3 Measuring Marriage and Fertility

Two of my central outcomes— Age at first birth and completed fertility—require combining information across census years. I describe their construction here.

Marriage To measure marriage over time, I focus on measures of never married in the 1930 and 1940 census. I also explore other marital status outcomes, like widowhood, divorce, and married with or without a partner present. The advantage of focusing on never married is that it captures the extensive margin of whether a woman has ever been married.

Completed fertility I measure completed fertility from own children observed in the household in 1930 and 1940. To do this, I restrict the analysis to women I observe in both the 1930 and 1940 censuses. I count the number of distinct own children a woman is observed with across the 1930 and 1940 censuses. I identify distinct children from their names and birth years so that a child appearing in both years is counted once.

I limit to women whom I observe in both 1930 and 1940, as measures of fertility from any single census risk miscounting the number of children ever born due to new children entering the household as older children may leave. This approach helps overcome this entry versus exit issue with children. However, this measure has limitations. It does not recover children who have died, were born and exited the household before 1930, or who were born after 1940. By 1940, women are on average 38, which is close to, but not complete, fertility (Appendix Figure B6 plots the age distribution of applicants in 1940).

Childless I define women as having no children if they are observed in both the 1930 and 1940 census and observed with none of their own children in their household. As referenced above, by 1940 women are approximately 38, which, while too young to be complete fertility, is a reasonable age to assume that women will not have their first child after this age.

Age at first birth I define women’s approximate age at first birth among women I observe in both 1930 and 1940 with a child. I identify the oldest child of all children reported in either the 1930 or 1940 census. I define women’s age at first birth as her age (in the census in which the oldest child is first observed) minus the age of the oldest child.

3.4 Data on Marriage Bars

I supplement my analysis with data on the geography of teacher marriage bars. To build this, I first use the cities in the US that have a marriage bar as reported in Yang and Brough (2026). Because their paper tracks the existence of the marriage bar in select large cities (most not in Illinois), I adopt their data-collection approach to the Illinois context. Their original data identified 18 distinct Illinois counties that had cities with the marriage bar. My expanded data build recovers an additional 10 counties in which there were reports of cities adopting the marriage bar. While there is no systematic data source to check the prevalence of the marriage bar, a 1942 survey of education in Illinois found that approximately one-third of reporting school districts operated marriage bars. This is not inconsistent with my finding of 28/102 counties being active marriage bar counties.

3.5 Record Linkage

3.5.1 Matching Back to Baseline Censuses

I match each examination candidate to the most recent full-count census preceding her exam—the 1910 census for exams taken in 1916–1919, and the 1920 census for exams taken in 1920–1923. In cases where I don’t find a match in the 1920 census for the 1920–1923 cohort, I attempt to match to the 1910 census. I use individuals’ first name, middle initial, last name, year of birth (inferred from reported age at exam), and residence as identifiers. I restrict the pool of candidate census records before matching to women enumerated in Illinois (excluding Cook County) in the 1910 or 1920 census.

Deterministic and Probabilistic Matches. I begin with a modified deterministic matcher that follows the ABE–Extra Information (ABE–EI) methodology (Abramitzky et al., 2025). The matcher cascades through match rounds that become progressively permissive based on name agreement. I start with exact name matches, and then move to phonetically standardized (sound-adjusted) names, and then to names that agree closely under the Jaro–Winkler string-similarity metric. In every round, I use extra information—middle initial and county of residence—to help distinguish (create uniqueness) among same-name candidates and require uniqueness within a 2-year birth-year window. I keep a candidate whenever a round produces a unique link. Each round attempts only the candidates still unmatched by the stricter rounds before it.

I supplement the deterministic matcher with a probabilistic record linkage using the Python package `splink` (Linacre et al., 2022), which implements Fellegi and Sunter (1969). This approach is useful in settings where the set of available identifiers differs across observations, for example if data fields are missing, and where measures are continuous. When linking each examination candidate to the 1910 or 1920 census, I use first and last names, middle initial, birth year (inferred from age at exam), and the geographic distance between her county of examination and her census place of residence. This differs from the deterministic approach, which matches based on county exactly (as opposed to geographic distance). To limit false links, I accept a pair only when every non-missing field agrees to within a set tolerance, and the link is one-to-one across both files. I only keep best matches whose match probability exceeds 0.90. In cases where deterministic and probabilistic matchers disagree, I keep the deterministic matches. Together, these procedures link just over 70 percent of exam candidates to a baseline census record.

3.5.2 Matching Forward to Outcome Censuses

To observe post-exam outcomes, I link women forward across censuses using the Census Tree crosswalks (Buckles et al., 2025), a set of high-coverage links between the full-count censuses constructed from a combination of machine-learning matching and crowd-sourced genealogical records. The Census Tree project is unique and necessary for this project because it matches women across census waves. Since my records only have maiden names, I rely exclusively on these genealogy-based links to track women into marriage.

Two concerns arise with forward matching. First, the pool of women who successfully match to future censuses may differ from the full exam-taking population. Matched women are somewhat more likely to have taken the exam in earlier years, to reside in rural areas, and to have passed the exam. To address this, I construct inverse probability weights (IPW) so that the women who match to subsequent censuses resemble the baseline matched pool on observables (estimates using this IPW, and other exercises are discussed in Appendix B.1). Second, the match rate may differ by treatment status. Reassuringly, I do not see that this match changes discontinuously at the RD threshold, as discussed in Section 5.3.

4 Sample Construction and Descriptives

In this section, I describe my main sample and characterize exam-takers' individual and childhood circumstances, and how they compare to other groups of similar-aged women in Illinois. This exercise helps us assess the generalizability of my population.

4.1 Sample Construction

First, I start with 48,029 unique applicants between 1916-1923. Then I restrict my sample to applicants where I observe 13 valid exam scores (35,517), who are women (28,180) and who are aged 17-25 at the writing of their first exam (26,367). My match-rate to the baseline census is 70%.

4.2 Exam Takers Compared to Similar-Aged Women

Table 1 compares exam applicants' childhood-household characteristics to several comparison groups: Normal School applicants, general Illinois women age 25 and under, and a pool of women who were eligible to sit the exam based on education.

Relative to Illinois. Relative to the general Illinois population (columns 2 and 6), exam applicants look more rural and lower-SES on household-head occupation score.

Relative to Illinois, Excluding Cook. This comparison, however, is driven almost entirely by Cook County (which includes Chicago): once Cook is excluded (columns 3 and 7), overall differences decrease significantly. Teacher applicants and downstate Illinois women are nearly identical on household-head occupation score (0.37 points lower), though they remain significantly more rural (+11 pp) and more likely to come from farm households (+9 pp). Household heads are a bit more concentrated in jobs that require higher education (+1.2 points) and are slightly more likely to have professional jobs (+0.7 pp). They are also, unsurprisingly, significantly more likely to be enrolled in school (+10 pp).

That teacher applicants are not more different from downstate women is somewhat surprising. To be eligible to teach, women had to complete at least 10 years of schooling (in practice, applicants had on average 11 years of schooling), putting them in the top half (based on 10 years) or top third (based on 11 years) of the education distribution for their birth cohorts.

Relative to Exam-Eligible Groups. Therefore, a second, potentially more natural comparison is to the subset of similar-aged women in Illinois whose reported completed schooling in the 1940 census is at least 10 years (i.e., women who would have been eligible to sit the exam). Because linking 1910 records forward to 1940 is selective, I weight this group by the inverse probability of matching to 1940, estimated from 1910 characteristics to make the matched sample representative of the underlying 1910 population.¹² Against this group of eligible women, exam applicants are much more *negatively* selected on socioeconomic status. Their household heads have occupation scores 2.6 points lower, are 18.4 percentage points more likely to be farmers than the eligible pool, and have heads of households in occupations that require less schooling (-3 points).

Together, these facts suggest that the exam-based teaching system drew highly educated women disproportionately from more modest, rural households. This made them, somewhat more comparable to non-Cook Illinois women as a whole than expected, but negatively selected relative to non-Cook Illinois women whose education levels would have made them eligible to be teachers.

¹²Defining eligibility from 1940 schooling is mildly post-treatment for the applicants who reach ten years partly because of teaching; since the comparison is on predetermined childhood-household characteristics rather than on schooling itself, this does not drive the pattern.

Relative to Normal School Enrollees. Finally, I compare exam applicants to women who pursued a teaching credential through enrollment at a Normal School. To build this comparison group, I digitize enrollment rosters from the Illinois State Normal School covering 1912–1923, which record roughly 4,000 enrollees; I link 2,390 of these women to the 1910 census.

Table 1 reports childhood-household characteristics for both groups (columns 5 and 9), where the difference column is a coefficient on a teacher-applicant indicator from a regression controlling for birth-year fixed effects. Exam applicants and Normal School attendees are similar in age and race. Exam applicants have household heads whose occupation scores are 1.7 points lower than those of Normal School attendees (on a base of 19.9) and are 9.1 percentage points more likely to be farmers (on a base of 40.0%); exam applicants are also somewhat less urban, by 9.4 percentage points (relative to 29.5% among Normal School attendees). Normal School students are also more than twice as likely (3.5 pp on a base of 2.7%) to have household heads in professional work and have household heads whose education score is 5 points higher, relative to exam applicants. This pattern is consistent with the exam functioning as a lower-cost entry point into teaching for rural, lower-SES women, relative to Normal School attendance, which required tuition, travel, and an extended residential commitment away from home.

5 Methodology

5.1 RDD around the First-Attempt Cumulative Score

A candidate passed if her cumulative score across the thirteen subject exams reached 975, an average of 75 percent. My RDD exploits this cumulative point cutoff to compare women who scored just above this threshold.

I measure each candidate’s score from her first exam attempt, since the carry-over of passing subject scores across retakes leaves later-attempt scores non-comparable across candidates. This follows the logic of GED research designs that avoid using retake outcomes as identifying variation because retaking is endogenous (Jepsen et al., 2016).

I estimate the following equation:

$$Y_i = \alpha + \tau \cdot \mathbf{1}(X_i \geq 0) + \beta_L X_i \cdot \mathbf{1}(X_i < 0) + \beta_R X_i \cdot \mathbf{1}(X_i \geq 0) + \varepsilon_i \quad (1)$$

Here, Y_i is an outcome of interest, for example whether woman i is observed as a teacher in a subsequent census; X_i is the running variable, measuring the distance of her first-attempt cumulative score from the 975 passing threshold (recentered to 0); τ , the parameter of interest,

captures the discontinuous jump in the outcome at the threshold; and β_L and β_R capture the slope of the regression function below and above the threshold, fit using a non-parametric local linear function. The main specification includes no additional controls. Because not all passers ultimately taught and some who failed found other routes into teaching, τ is an intent-to-treat effect of crossing the passing threshold.

I implement the bias-corrected estimator of Calonico et al. (2017), using a local-linear specification (polynomial order one) with a triangular kernel and the MSE-optimal bandwidth. I use a triangular kernel throughout because it down-weights observations far from the threshold, which limits the volatility that can otherwise arise in small-sample RD estimates. In Appendix B.1, discussed in Section 9, I show that my results are robust to these and other specification choices as well as to the inclusion of covariates.

The design relies on two key assumptions. First, that there is continuity in the running variable. Several features support this. First, because candidates were scored from 0-100 on 13 exams, my scores range from 0-1300, which is a relatively continuous distribution compared to a more discrete setting (for example, integers between 0-100). Second, candidates wrote anonymized exams that were graded centrally, leaving little scope for one to manipulate her own score. However, there is still the case of grader manipulation, which I address in Section 5.2.

Second, absent passing, the outcome must evolve smoothly across the threshold. While this continuity assumption is untestable, I assess it indirectly by testing whether predetermined characteristics change discontinuously at the cutoff (Section 5.3). The balance table (Table 2) serves both as a diagnostic and as a useful summary statistics table.

5.2 Identifiable grade bumping and implications for smoothness

I identify one key, but addressable, threat to the RD continuity assumption. Recall that in principle, applicants were required to meet both the average threshold (975) and a minimum threshold—specifically, that no single subject of 13 can fall below 60. This created incentives for examiners to selectively bump one’s minimum subject score to exactly 60 for candidates for whom this change would allow them to pass the exam. Because these bumps added points only when they carried a candidate to or beyond the passing margin, they generated excess mass at just above the 975 cutoff, a corresponding missing mass below the 975 cutoff, and a discontinuous break in the density of first-attempt scores. However, this can be corrected by removing individuals whose minimum score is exactly 60.

To observe this, see Figure 4, panel (A). Here, the histogram plots the number of applicants at each single-cumulative score relative to the 975 threshold (centered to 0). Those in

grey represent individuals whose minimum exam score across the 13 is a value other than 60. Those shaded in purple represent individuals whose minimum exam score is exactly 60.

The counterfactual curve (plotted in green) fits a smooth skew-normal distribution of cumulative scores to the data. It is clear that there is significant excess mass occurring just to the right of the passing threshold, which extends relatively far to the right side of the distribution, and a hollowing of mass to the left. In the full sample, the Cattaneo et al. (2018) density test strongly rejects the null of no manipulation ($p < 0.01$, with t -statistics exceeding 5, panel C). However, once I remove individuals whose minimum score is exactly 60, the distribution appears smooth (panel B). Importantly, this exclusion is symmetric around the 975 cutoff, meaning that the interpretation of the RD is unchanged and the design still compares those who marginally passed versus failed to meet the 975 cumulative-score cutoff, among the applicants who remain. This does not break continuity in the running variable, since all cumulative scores around the cutoff are represented regardless of whether we include individuals whose minimum score is 60.¹³ Using the Cattaneo et al. (2018) density estimator, I cannot reject density smoothness ($p = 0.42$).

To interpret what kind of manipulation generates this pattern, I simulate several simple grading regimes in Appendix Figure B7. The data are most consistent with a regime in which candidates first receive grades on all thirteen exams. Then examiners bump sub-60 scores to exactly 60 only for candidates for whom this change would allow them to pass the exam (panel A). This regime produces a density break in the full sample—excess mass just to the right of the cutoff, with a smaller, more spread-out shortfall to the left of it. This is because the regime increases the cumulative scores of individuals only when doing so lets them meet the average. However, this excess mass is identifiable and can be excluded from the distribution by dropping observations with a minimum subject score of exactly 60.¹⁴

By contrast, direct manipulation of the average score at the cumulative cutoff (panel B, left) results in a narrow excess mass occurring exactly at the pass threshold, with a corresponding “hollowing” in the distribution to the left of the cutoff. Excluding 60s has no corrective value in this regime. Similarly, a regime in which the probability that one’s minimum is bumped up to 60 is distributed uniformly (i.e., not targeted to passers) (panel B, right) is also inconsistent with the data. In this regime, everyone’s grades are uniformly higher in a way that produces no excess mass at the pass threshold, and excluding those

¹³This is distinct from a donut design, which explicitly breaks continuity in the running variable.

¹⁴Though, given heaping at the value 60, the newly recovered distribution is below the unmanipulated distribution. Heaping in the exam scores is evident in Figure B8, in which it is clear that 60 is the most common single grade. Figure B9 decomposes this further, highlighting that when one removes individuals whose minimum is exactly 60, this removes the excess mass (manipulation)—the region shaded in purple—but it also excludes the additional people who obtained a minimum of 60 without manipulation—the region shaded in gold.

with a minimum of 60 does not change the shape of the distribution.

I also show that my results are robust to including the full population of test-takers, including those who score exactly a minimum of 60. Because retaining these observations violates the continuity assumption underlying my preferred local-polynomial estimator (Calonico et al., 2017), these checks instead rely on alternative RD estimators whose identifying assumptions do not require a smooth density of the running variable at the threshold. Specifically, I show robustness to using a local randomization design (Cattaneo et al., 2015) which does not rely on the smoothness assumption but instead imposes stronger assumptions around the quasi-random assignment to either side of the cutoff (which tends to require small bandwidths).

5.3 Summary Statistics and Covariate Balance

First, I find no evidence that women who marginally meet the cumulative score criteria differ from marginal failers in urbanicity, age, or household head occupational scores. Reassuringly, I also do not find any difference in probability of matching to subsequent censuses; the by-census estimates range from -0.029 to +0.03, and the pooled measure of “matched to any subsequent census” is +0.013 ($p = 0.70$). The one possible extension is that marginal passers appear to be more likely to be living in their childhood household in their baseline census (observed in 1910 for exams taken between 1916-1919, and in 1920 for exams taken between 1920-1923). However, these magnitudes are relatively small (+3 pp on a baseline of 0.90-0.93).

This table also helps characterize the marginal population, among whom the effects of access to teaching will be identified. On average, she is young (age 18.2) at the time of her exam and has completed 11 years of schooling. She is disproportionately from a rural household (80%) and a farm household (48%).

5.4 First Stage Effects

Figure B11 plots the relationship between meeting the cumulative score on one’s first attempt and the probability of ever clearing the average for my main estimating sample. Passing the average on the first attempt raises the probability of ever passing by about 50 percentage points (robust $p < 0.001$). Among all women who ever clear the average, roughly 60 percent do so on their first attempt. Throughout, I present ITT estimates; however, to scale the presented ITT estimates accordingly, it is appropriate to multiply by 2 to recover the LATE.

6 Labor Effects

In this section, I first investigate the relationship between marginally passing and women’s employment as teachers as well as their labor force participation as a whole. I also investigate income, which is observed only in 1940.

6.1 Teaching

Figures 5 and 6, together with Table 3, report the immediate and long-run effects on teaching. First, I show that marginally passing the certification exam increases the probability of being observed in subsequent censuses as a teacher by 13.6 p.p., or 36% on a baseline of 38% (Table 3, Column 1).

This estimate is strong evidence of a large first stage of passing the certification exam on teaching. However, the magnitude needs careful consideration. The decennial nature of the census means that the full trajectory of women’s labor market outcomes is not observed. Ideally, whether a woman was ever employed as a teacher would be observed on an annual basis, whereas the decennial censuses provide only snapshot views in 1920, 1930, and 1940. This measure misses, for example, women who pass the exam in 1923, teach for several years, but exit teaching before the 1930 census.

To investigate how women’s teaching evolves over time, Figure 6 plots the effects of certification on teaching in the short, medium, and long run.

To identify the short-run effects, I focus on exam cohorts 1916-1919 and their observed teaching status in 1920. Passing increases the probability of reporting a teaching occupation in the 1920 census by 18.6 percentage points ($p < 0.01$; Table 3, Column 3). The response is even larger among the recent 1918–1919 cohorts, whom I observe approximately one to two years after the exam. Passing raises teaching by 21.4 percentage points ($p < 0.01$; Column 2).¹⁵

Over the long run, this initial teaching effect dampens but remains economically meaningful. Passing raises teaching by 8.0 percentage points in 1930 ($p = 0.02$; below-cutoff mean 24.0 percent; Column 4) and by 8.5 percentage points in 1940 ($p = 0.01$; mean 14.1 percent; Column 5). Pooling the 1930 and 1940 estimates, marginal passers are 11.1 p.p. (44%) more likely to be teachers in the long-run (Table 8, Panel B).

¹⁵Note, while 1920 is the best census year to observe short-run effects, it has several limitations. First, there was a teacher shortage in 1919, leading to a large number of emergency certificates being issued in 1919 (40% of elementary credentials issued in 1919 were on an emergency basis, which is approximately double that of typical years). This may lead to an inflation in the number of non-passing teachers observed as teachers in this census. As such, I view 21 percentage points as a likely lower bound on the immediate short-run effects, but it is the best approximation given data constraints.

This decay rate is unsurprising given general teaching retention in this period (Appendix Figure B12). Among women who are teachers in 1920, 34% are still teachers in 1930, and 29% are still teachers in 1940. Thus, while there is significant exit from teaching in the first decade, conditional on still teaching in 1930, the large majority remain teachers in 1940.

6.2 Labor Force Participation

Passing also increases extensive-margin labor force participation. Overall, marginal passers are 9.6 percentage points (16%) more likely to ever be observed in the labor force ($p = 0.005$; below-cutoff mean 59.0 percent; Table 4, Column 1). This is not an obvious corollary of the teaching estimates. Women who fail could have entered other occupations, but instead it appears that the majority of women on the margin of teaching were also on the margin of labor force entry. Consistent with this, the labor-force effect is not statistically distinguishable from the teaching effect (a difference of 3.8 percentage points, $p = 0.13$; Appendix Table A8).

As with the teaching effect, I decompose this LF effect over time. Figures 7 and 8, together with Table 4, show an increase of 19.3 percentage points among the 1918–1919 cohorts and 16.6 percentage points among all 1916–1919 cohorts in 1920 (both $p < 0.01$; Columns 2 and 3). Over the longer run, the participation effect is 7.3 percentage points in 1930 ($p = 0.04$; below-cutoff mean 40.0 percent; Column 4) and 8.9 percentage points in 1940 ($p = 0.02$; mean 30.2 percent; Column 5).

The participation gains also translate to earnings. Figure 9 and the final column of Table 4 (Column 6) show that passing raises a woman’s own wage income in 1940 by \$76.3 ($p = 0.08$), approximately 28 percent of the below-cutoff mean of \$273; this is driven entirely by extensive margin LFP.

6.3 Labor Force Persistence

Marginal passers remain more likely to be in the labor force and to remain teachers both in the long run—7 to 14 years after the exam—and in the very long run—17 to 24 years after the exam. In this section, I examine factors driving this persistence, focusing on teaching and labor force participation in the 1930 and 1940 censuses. Because both censuses capture the same long-run horizon, I pool them where each outcome equals one if a woman is observed in the indicated state in either 1930 or 1940. For example, the outcome “married teacher” equals one if a woman is observed teaching with a spouse present in either census. Appendix Table A5 reports the estimates separately by census wave.

6.3.1 Teaching Drove Persistence

I first ask which occupations account for the long-run labor force response (Table 8). Pooling the 1930 and 1940 censuses, passing raises labor force participation by 8.9 percentage points (19 percent; Panel A, Column 1), and the main driver of this response is teaching (Panel B, Column 1). Passing has no detectable effect on non-teaching white-collar work or on other work (Panels C and D), which makes teaching account for essentially all of the long-run participation response, though this does differ somewhat by marital status, as discussed next. Figure B4 shows the shares of the top 10 occupations observed in the full sample in the 1930–1940 censuses. The leading occupation is teaching at 52.9%, followed distantly by stenography (8.1%) and bookkeeping (5.4%).

6.3.2 Married Women Persisted

I next ask whether the long-run effects are driven by married or unmarried women. Because my main estimates in Table 8 are pooled outcomes across the 1930 and 1940 censuses, and because married women’s marital status can change between censuses, the married and unmarried cells overlap, and their estimates do not sum to the totals due to pooling and to different bandwidths. For census-wave-specific estimates, see Appendix Table A5.

Passing raises the probability of non-married LFP by 6 percentage points (17%) and married LFP by 5 percentage points (42%) (Table 8, Panel A, Columns 2 and 3). For unmarried women, this LFP effect is driven by teaching (+9 pp; Panel B, Column 2), which potentially crowded out white-collar work (−3 pp, $p = 0.12$; Panel C, Column 2). In contrast, married women’s LFP effect is driven mostly by teaching (+3 pp; Panel B, Column 3) and potentially by other white-collar work (+1.8 pp; Panel C, Column 3).

These pooled effects mask some heterogeneity by census wave (Appendix Table A5), in which married women’s LFP effects in 1930 are driven by other white-collar work, and not at all by teaching, whereas by 1940 married LFP is driven exclusively by teaching. In 1930, single LFP is driven by teaching with some crowd out of other white-collar work, and in 1940 exclusively by teaching. These findings suggest that, while teaching may have some medium-run effects, its long-run impacts on LFP are driven by those who teach.

To better understand the composition of those in the LF and as teachers in 1940, Figure 14 plots the entry and exit patterns of women relative to their 1930 occupation/LFP. I plot this for the full sample, as the RD is not suited to identify granular transition behavior, and so this exercise is interpreted as descriptive not causal. I see that 9% of teachers in 1940, and 19% of women in the LF in 1940, are women who were married and not teaching (respectively, not in the LF) in 1930 and who are still married in 1940, suggesting reasonably large rates

of re-entry into the profession—and even more so into the labor force—for married women. This suggests that, while for many teaching was a sustained and continuous career, acquiring a teacher certificate in early adulthood may have allowed women to enter, exit, and then re-enter the LF later on. This is consistent with the observation that 1940 and 1950 married LFP was driven by older married women, precisely those with past work experience (Goldin, 1989).

6.3.3 Education Reinforced Persistence

Next I investigate how labor force persistence interacts with human capital accumulation. In Table 9 (Columns 1–2), I show that marginal passers are slightly more likely to acquire additional education after the exam (+0.2 years, $p = 0.11$) and significantly more likely to have 14 or more years of schooling (+9 pp). Acquired education is defined as the number of years of schooling in the 1940 census minus the number of years of schooling observed at the time of the exam.

To understand who drives these additional education effects, I multiply acquired education by LFP status in the 1930 and 1940 censuses. I see that the entirety of the education effect is driven by women who, in 1930 or 1940, are teachers (estimate of 0.25 years, $p = 0.035$; Column 5), with little to no contribution among women not observed in the labor force in either census (−0.01 years, $p = 0.93$; Column 3) and among women observed in the labor force but not as a teacher (−0.06, $p = 0.38$; Column 4).

These joint outcomes do not establish that education causally produces persistence, but they suggest that education and labor force persistence are complementary.

7 Family Effects

I then turn to family outcomes where, unlike the teaching or LFP outcomes, never-married and fertility measures are absorbing in nature, which facilitates better measurement of these outcomes. This also means, for example, that one should not try to ‘scale’ family outcomes by LFP or teaching outcomes, since the LFP and teaching measures are incomplete. Finally, I examine whether changes in income translate to changes in household resources.

7.1 Marriage

Access to teaching also changed women’s marriage outcomes. Table 5 and Figure 11 show that passing raises the probability of never having married by 1930 by 8.5 percentage points ($p = 0.02$), against a below-cutoff mean of 34.8 percent (Column 1). The effect attenuates

to 5.0 percentage points by 1940 ($p = 0.07$; mean 19.3 percent; Column 2), which displaces marriage with a spouse present (which falls by 6.3 p.p., $p = 0.04$; Column 3). Divorce and widowhood are unchanged (Columns 4 and 5). This pattern is consistent with at least part of the response operating through delayed marriage.

7.2 Fertility

Next, I consider completed fertility (Table 6 and Figure 12). To analyze fertility, I limit the sample to women observed in both the 1930 and 1940 censuses. Because the census does not record children ever born,¹⁶ I measure fertility as the number of distinct children ever observed in a woman’s household across the two waves. Limiting the sample to women who appear in both censuses allows me to capture children who appear in 1930 but have left the household by 1940, and children who arrive in 1940 but were not present in 1930. Section 3 provides details. On average, women are 38 years old in 1940, so these measures are close to, but likely not exactly, completed fertility.

Overall, marginal passing lowers estimated fertility by 0.35 children ($p = 0.02$), about 17 percent of the below-cutoff mean of 2.07 (Column 1). This effect is driven by both the extensive and intensive margins of fertility. The probability of having no children rises by 7.1 percentage points ($p = 0.03$; Column 2). At the same time, fertility conditional on being positive falls by 0.23 children ($p = 0.22$; Column 3). This second effect is likely driven by a delay in family formation. The point estimates therefore suggest contributions from both childlessness and family size among mothers, although each component is less precisely estimated than the total effect. Passing also delays age at first birth by 0.76 years ($p = 0.06$; Column 4).

7.3 Women’s Contribution to Household Wage Income

Table 7 shows a positive but statistically insignificant effect on total household wage income (\$80.0; $p = 0.17$; Column 1). This increase is driven by women’s own earnings. Passing raises a woman’s own wage income by \$64.2 ($p = 0.10$; Column 2),¹⁷ concentrated among women without a co-resident spouse (\$51.4; Column 5) rather than among those with one (\$12.8; Column 4), though neither component is individually precise.

Husband-contributed wage income contributes little (\$8.3; $p = 0.61$; Column 3), masking a positive effect among married women and a negative effect among unmarried women, so

¹⁶The 1940 census asked children ever born only of the sample-line subsample.

¹⁷This estimate uses a common bandwidth across the household-income components and so differs from the \$76.3 in Table 4, which uses the outcome-specific MSE-optimal bandwidth.

the household total income, though imprecise, reflects the rise in women’s own earnings rather than any offsetting change in husband income. Beyond earnings, I find no evidence that passing changed whom women married; husbands are no better matched on education, occupational standing, or income rank relative to the woman’s father (Appendix Table A3). Appendix Table A1 shows no discontinuity in income-data availability.

8 Heterogeneity

While my sample is not large enough to support extensive heterogeneity analysis, I present suggestive analyses describing how effects differed across locations with and without the marriage bar, and across women from farm versus non-farm households. In each heterogeneity analysis, I estimate group-specific RD effects with jointly chosen, group-specific bandwidths and test the statistical difference across groups using [Calonico et al. \(2025\)](#). While no individual subgroup difference is statistically significant at the 5 percent level, I emphasize the pattern of point estimates.

8.1 By Marriage-Bar County

Whether married women could keep working depended in part on whether they lived in places that barred married women from teaching. Many districts operated such marriage bars, either formally or informally (Section 2). Figure 13 splits the married-women employment effects by whether a woman’s county of residence has a documented marriage bar. The married-teacher effect is concentrated in counties with no documented bar; in counties with a documented bar, it is close to zero. There is no evidence that married women in bar counties shifted into other forms of work; their effects on non-teaching white-collar and other work are small and similar to those in counties without bars.

Appendix Table A7 extends this heterogeneity analysis to the full outcome set. The point estimates suggest the family response also differed by bar status. Extensive-margin effects are often larger (ever married by 1930 falls 15.1 percentage points; childlessness rises 11.2 points) in locations with marriage bars, whereas intensive-margin effects (first births are delayed by 1.6 years) are larger in locations without the marriage bar.

8.2 By Farm vs. Non-Farm Household

Roughly half of fathers are farmers, who share a single occupation score. Therefore, instead of splitting my sample by household occupation score, the most appropriate heterogeneity

analysis by childhood characteristics is whether a woman’s father was a farmer, as measured in her 1910 childhood household (Section 3).¹⁸ The split is reported in Appendix Table A6.

I find that passing increases teaching similarly in both groups (1940: +9.7 percentage points farm, +8.9 non-farm; both $p < 0.05$). However, labor force effects, and other downstream outcomes, are consistently larger among farmers’ daughters, who had the fewest local white-collar alternatives. Own wage income rises \$168 ($p < 0.01$) among farm daughters against $-\$10$ ($p = 0.92$) among non-farm daughters; long-run labor force participation, schooling (+0.69 years, $p = 0.01$, vs. +0.13), and the fertility decline (-0.51 children, $p = 0.01$, vs. -0.17) are all concentrated in the farm group. For non-farm daughters, who more often had access to other white-collar work, passing shifted women into teaching without an accompanying rise in labor force participation. This is consistent with teaching playing a more central role in women’s lives when outside options were scarce, and provides a new context to study women’s work relative to existing work, which focuses on clerical work or relies disproportionately on white-collar shocks in urban settings.

9 Robustness

I show robustness of my results to several different empirical design decisions. The results of these exercises are reported across Appendices B.1 and C.

Fixed bandwidths: The main estimates use MSE-optimal bandwidths chosen outcome-by-outcome. I re-estimate every outcome at common fixed windows around the cutoff (25, 50, and 75), so that differences across outcomes cannot reflect differences in the estimation sample.

Donut: To ensure estimates are not driven by single individuals exactly at the threshold, I drop observations in the 11 score points nearest the cutoff and re-estimate on the remaining sample.

Kernel weighting: The main estimates use a triangular kernel; I re-estimate with a uniform kernel, which increases the weight given to observations that lie further from the cutoff.

Alternative SE construction: I report conventional, robust bias-corrected, and alternative clustering choices for the main outcomes.

Local randomization including individuals scoring 60: As an alternative to continuity-based estimation, I use local randomization inference (Cattaneo et al., 2015) in a fixed 13-point window, including examinees whose minimum subject score of exactly 60 placed them at the

¹⁸Taking the father from the 1910 household, recovered via the woman’s own childhood census record, raises coverage to about 93 percent of the sample; the split is close to even (49.7 percent farm).

administrative floor for admission. I use a smaller window because the identifying assumption in local randomization is that assignment to either side of the cutoff is as good as random within a bandwidth, and therefore favors narrow bandwidths.

Including covariates: I add predetermined covariates (age at exam, years of education at exam, and birth year) to the main specifications; covariate adjustment should affect precision but not point estimates if the design is valid.

Placebo cutoffs: Finally, I re-estimate the main specifications at artificial cutoffs placed away from the true passing threshold; the full placebo distributions for every outcome are reported in Appendix C.

10 Aggregate Implications

In this section, I provide back-of-the-envelope estimates to translate the observed RD estimates into overall changes in LFP and fertility in Illinois (excluding Cook County). To do this, I define the share of an aggregate change explained as the number of women who additionally entered a career, multiplied by the effect per entrant, divided by the aggregate change itself. Because the regression discontinuity estimates apply to women near the certification threshold, I multiply my RD estimates by the size of that marginal population and compare the result to the aggregate changes in fertility and LFP over relevant age cohorts. I compute this only among women in Illinois, excluding Cook. This is a conservative approach since I do not assume anything about how the effect, identified on marginal women, translates to different contexts. However, given that many women in the relevant birth cohorts (11%) are observed taking the teacher certification exam, this exercise produces non-trivial aggregate estimates.¹⁹

In this exercise, I consider women marginal if their cumulative scores fell between 72 and 78 percent and if they passed, which made up 9.9% of test-takers. I count only the passers because my estimate is the effect of crossing the threshold. This then implies that roughly 1% of downstate women in the 1897–1902 birth cohorts were on the margin of access.

$$\underbrace{0.113}_{\text{sat the exam}} \times \underbrace{0.099}_{\text{passed within } \pm 3 \text{ of cutoff}} \approx 0.011,$$

I multiply this 1.1% by each reduced-form effect (1940 LFP: Table 4, Column 5; fertility: Table 6, Column 1) and divide the resultant by the corresponding aggregate change among

¹⁹19,805 unique women applicants born 1897–1902 in the exam registers, divided by 174,724 Illinois women in those birth cohorts in the 1920 full-count census, excluding Cook County.

downstate-Illinois women to estimate the share of each aggregate change attributable to entry at the certification margin.

I calculate the aggregate change in LFP among women aged 35–44, comparing the 1940 census to the same age group in the 1900 census. To estimate changes in fertility, I instead compute the difference in completed fertility among women aged 45–54 in the 1950 census relative to women aged 45–54 in the 1900 census.²⁰

For women aged 35–44 between 1900 and 1940, labor-force participation rose from 10.5 to 21.5 percent. For women aged 45–54, completed fertility fell from 4.83 children (1900 full count) to 2.67 (1950 sample-line children-ever-born, with never-married women assigned zero).

Under a very limited assumption approach (only that the effect is constant among women scoring 72–78 points), the certification margin accounts for a small but non-trivial share—roughly one percent—of the cohort rise in downstate-Illinois women’s labor-force participation, and 0.2 percent of the fertility decline.

$$\begin{aligned} \text{Female labor force (1940): } & \frac{0.011 \times 8.9 \text{ pp}}{21.5\% - 10.5\%} \approx 1\%, \\ \text{Completed fertility: } & \frac{0.011 \times 0.35 \text{ children}}{4.8 - 2.7 \text{ children}} \approx 0.2\%. \end{aligned}$$

To gauge what the estimates imply for white-collar work more broadly, I extend them to the era’s other new white-collar entrants.²¹ Among American women aged 18–24, employment in white-collar occupations rose from 3.4 percent in 1880 to 19.7 percent in 1920, accounting for the entire net growth in young women’s employment. Scaling my estimates to this population requires two changes. Because passing raised the probability of ever being certified by 50 percentage points, the effect per certified woman is twice the reduced form (17.8 percentage points on 1940 participation and -0.70 children). The ideal data would allow us to identify what fraction of women are induced to ever be employed as a teacher as a result of certification. However, in absence of this we can assume that likely-first stage is somewhere between the short-run teaching margin (21 percentage points)²² and the first-stage on certification (50 percentage points),²³ which imply an effect per entrant of 17.8 to

²⁰Fertility among women aged 45–54 is effectively completed fertility, measured as children ever born. The question was asked only of ever-married women, so never-married women are counted as zero. All samples are women in non-Cook County, Illinois.

²¹Appendix Table A9 reports the same calculation for all teachers, rural teachers, all professional occupations, and all white-collar work.

²²This is the 1920 effect for exam takers from 1918–1919.

²³This assumes that all women who gained teaching credentials entered.

41.6 percentage points on 1940 participation and -0.70 to -1.64 children. And because my estimates come from women whose alternative was not working, I count only the 61 percent of new entrants who came from non-employment.²⁴ Under the assumption that the persistence effects of early-career entry extend to these occupations, additional early-career entry accounts for roughly 16 to 36 percent of the rise in American women’s prime-age labor force participation between these cohorts, and roughly 4 to 8 percent of the national decline in completed fertility. The two bounds differ only in the first stage.

$$\begin{aligned} \text{Prime-age labor force: } & \frac{0.164 \times 0.61 \times 17.8 \text{ pp}}{27.1\% - 15.7\%} \approx 16\% \quad \text{to} \quad \frac{0.164 \times 0.61 \times 41.6 \text{ pp}}{27.1\% - 15.7\%} \approx 36\%, \\ \text{Completed fertility: } & \frac{0.164 \times 0.61 \times 0.70 \text{ children}}{4.71 - 2.76 \text{ children}} \approx 4\% \quad \text{to} \quad \frac{0.164 \times 0.61 \times 1.64 \text{ children}}{4.71 - 2.76 \text{ children}} \approx 8\%. \end{aligned}$$

11 Conclusion

In this paper, I provide estimates on the long-run effects of women’s access to professional work during a period of substantial change in women’s employment, education, and family formation. I exploit individual-level variation in access to teaching generated by the passing threshold on Illinois teacher-certification examinations. Using newly digitized examination records linked to the full-count census, I show that women who marginally gained access to teaching were significantly more likely to become teachers and to enter the labor force. These effects persisted for more than two decades. Marginal passers remained 44% more likely to teach and 19% more likely to participate in the labor force in the long run.

Women’s long-run persistence in the labor force was driven almost entirely by continued employment in teaching. A substantial share of the effect was also driven by women who continued working after marriage. Finally, marginal passers acquired more education after the exam, with these gains concentrated among women who remained teachers. Together, these findings suggest that access to teaching created sustained careers and encouraged further investment in education that reinforced LFP persistence.

Access to teaching also changed women’s family formation in both the intensive and extensive margins. Marginal passers delayed marriage, delayed childbirth, and were more likely to remain unmarried and childless, and had fewer children. Access to teaching did

²⁴The 61 percent follows from where the entrants came from: among women aged 18–24, non-employment fell from 71.4 to 61.4 percent and other gainful work from 25.2 to 18.9 percent. Restricting the exercise to downstate Illinois—the setting of my estimates—yields nearly identical results: white-collar entry rose from 4.2 to 18.9 percent. Within Illinois, the persistence rate is nearly identical whether women began as teachers (36.1 percent), other professionals (37.9 percent), or clerical workers (36.2 percent).

not, however, allow women to marry husbands with higher levels of income, education, or occupational standing; instead, the economic gains from teaching were concentrated among women themselves.

Extending these estimates to new entry to white-collar work, I show that additional early-career entry accounts for 16 to 36 percent of the rise in prime-age labor force participation and 4 to 8 percent of the decline in completed fertility.

These findings show that career women were not a predetermined group. Women could be induced into sustained careers when career opportunities became available. Where teaching was one of the few professional occupations open to women, access to it shaped not only whether women worked, but whether they kept working after marriage, continued their education, and formed families. Together with the aggregate estimates, these results suggest that early-career access to white-collar work was a key force in the transformation of women's economic and family lives.

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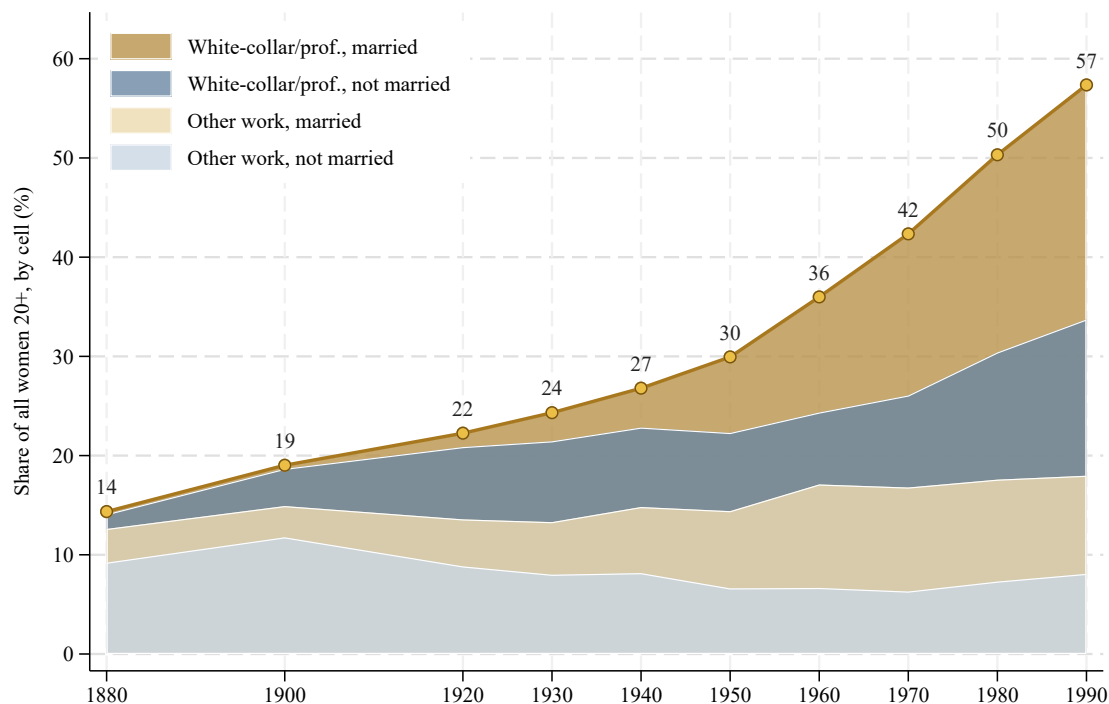
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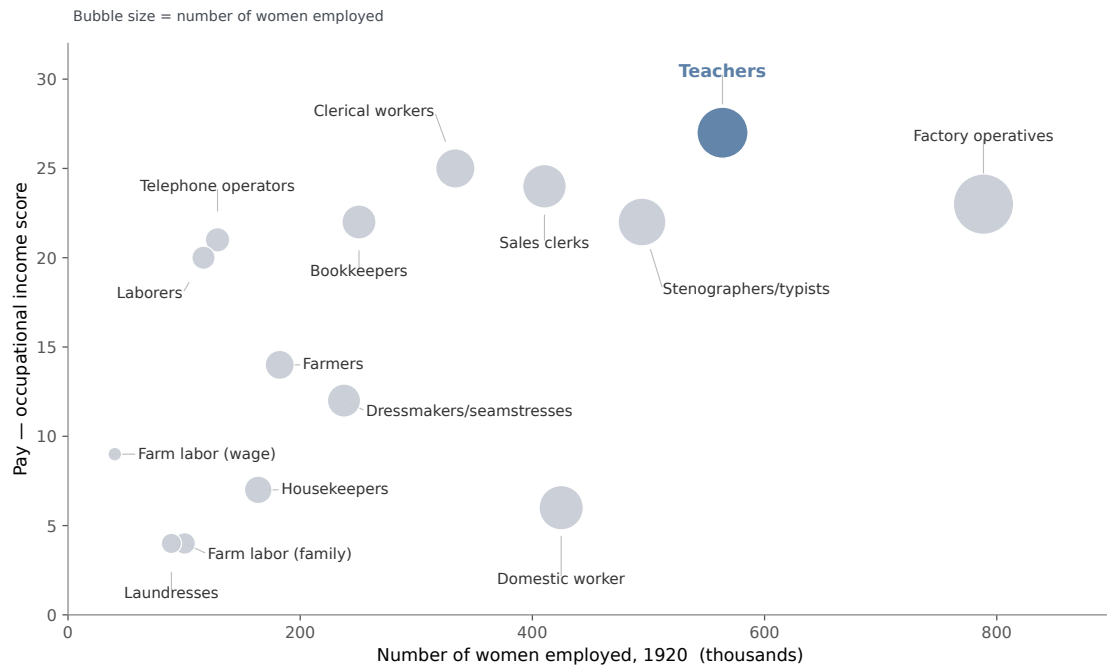
12 Figures

Figure 1: The transformation of women’s work, 1880–1990



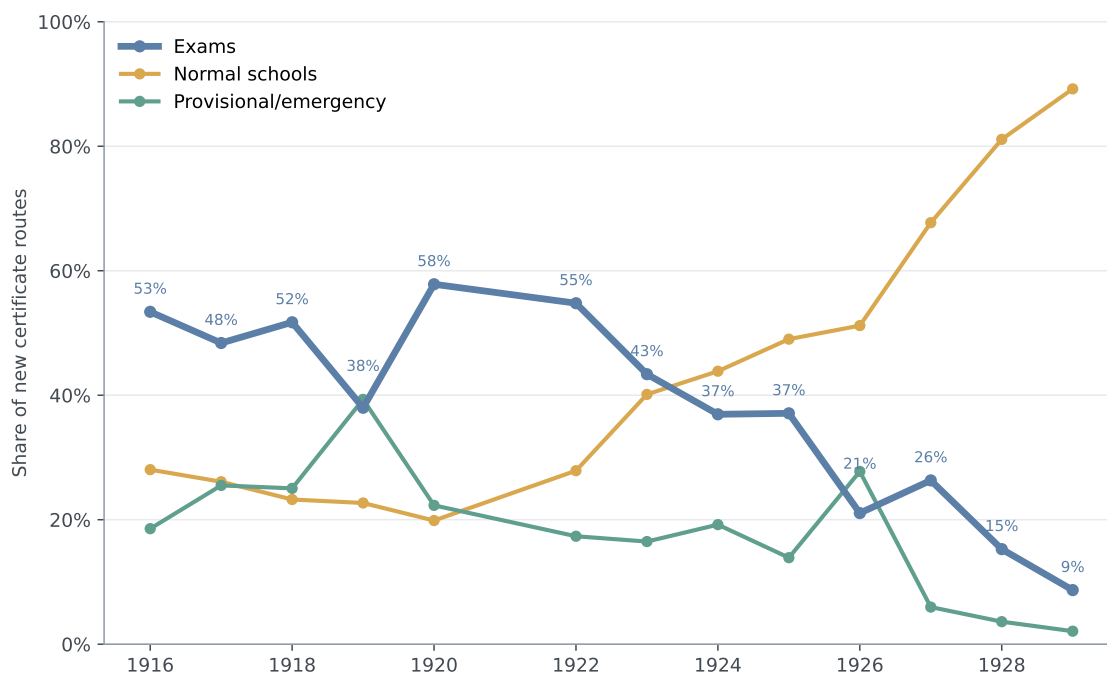
Notes: The figure decomposes female labor-force participation into four mutually exclusive groups defined by marital status (currently married versus not currently married) and occupation type (white-collar/professional versus other work). The gold marker line and value labels give total participation. White-collar/professional work is 1950 occupation codes 1–99 (professional and technical) and 200–490 (managerial, clerical, and sales). Data are the IPUMS 1% samples of the decennial censuses for 1880–1990 (the 1910 sample is excluded because its enumeration overstates women’s gainful employment); estimates are weighted by person weights. Appendix Figure B3 plots the same series for professional and technical work alone.

Figure 2: Teaching as a large, high-paying occupation for women



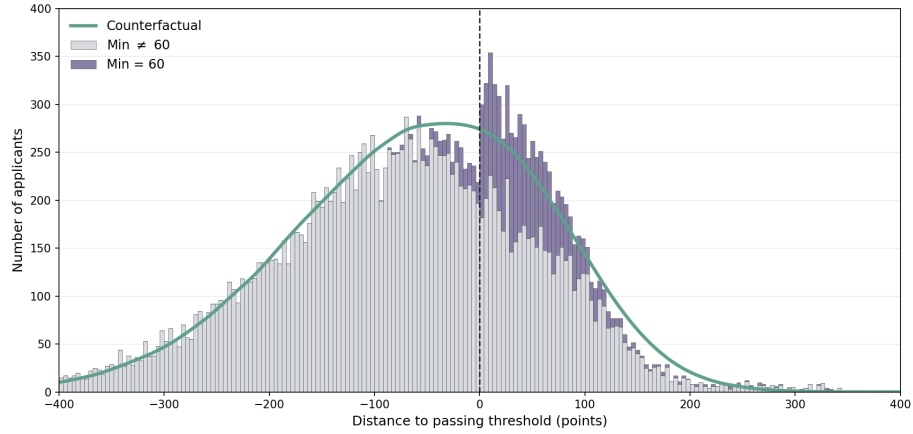
Notes: The figure plots major occupations for white women aged 19 and older in the 1920 Census labor force. The x-axis gives the number of women employed in each occupation, and the y-axis gives the occupational income score. Bubble size is proportional to the number of women employed.

Figure 3: Routes to elementary teaching certificates for women outside Cook County

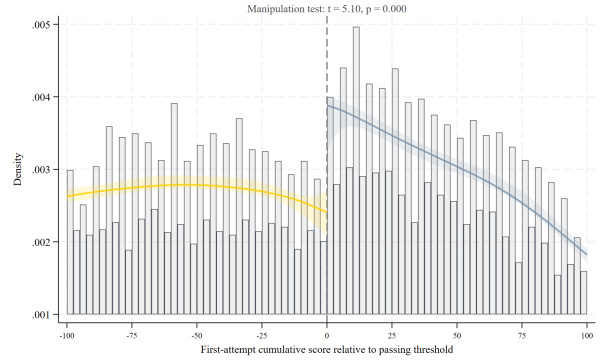
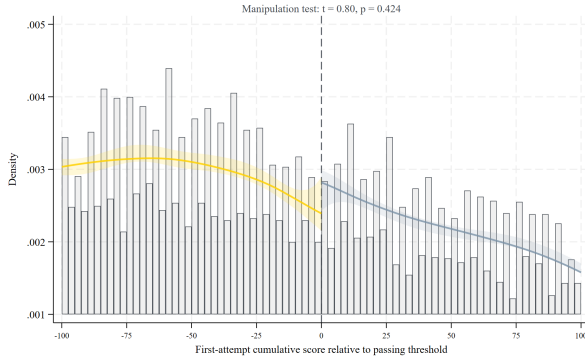


Notes: The figure plots new second-grade elementary certificate routes for women in Illinois, excluding Cook County. Normal schools are certificates issued based on credits filed. Provisional/emergency combines provisional and emergency certificates.

Figure 4: Recovering Continuity in Passing Threshold



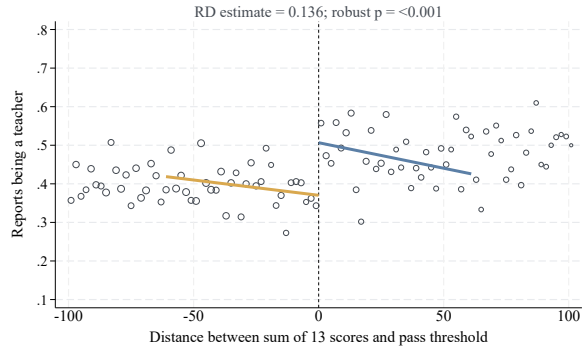
(a) Excess mass of exactly-60 records above the counterfactual



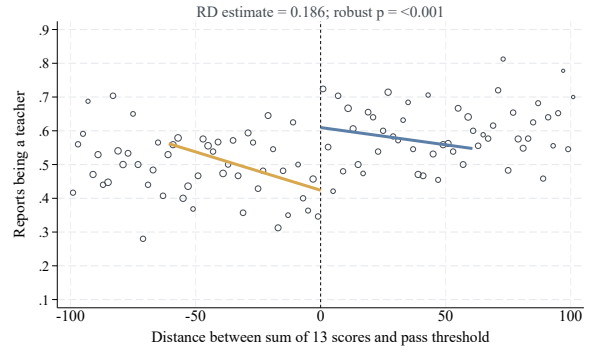
(b) Excluding bumped records (min score $\neq 60$) (c) Including bumped records (min score = 60)

Notes: First-attempt cumulative exam scores for all test-takers, recentered so the 975 passing threshold falls at 0. Panel (a): records with a minimum subject score of exactly 60 (purple) are stacked as excess mass above records with a minimum score not equal to 60 (grey). The counterfactual (green) is the unmanipulated distribution from the grading simulation in Appendix Figure B7, scaled to the same number of applicants. Panels (b) and (c) report Cattaneo et al. (2018) manipulation tests for test-takers aged 17–25: removing the exactly-60 records leaves the density continuous at the threshold ($t = 0.80$, $p = 0.42$), while including them produces a sharp discontinuity ($t = 5.10$, $p < 0.001$).

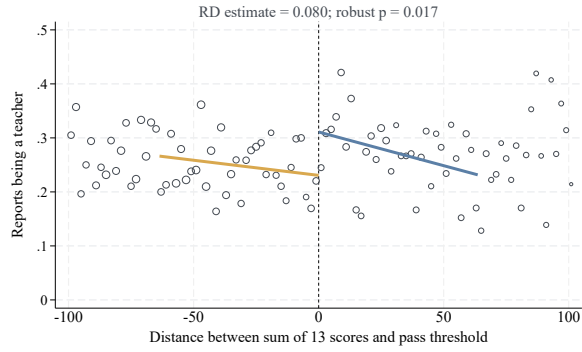
Figure 5: Effects on Teaching



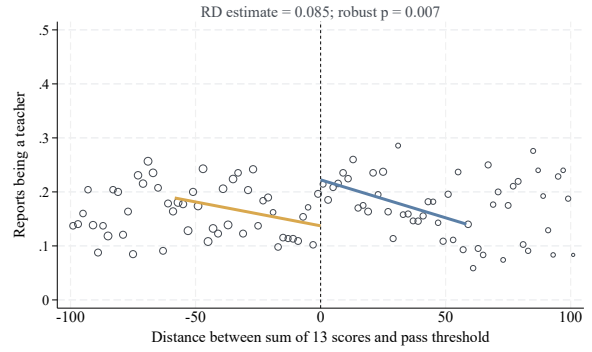
(a) Ever observed teaching by 1940



(b) Teaching in 1920



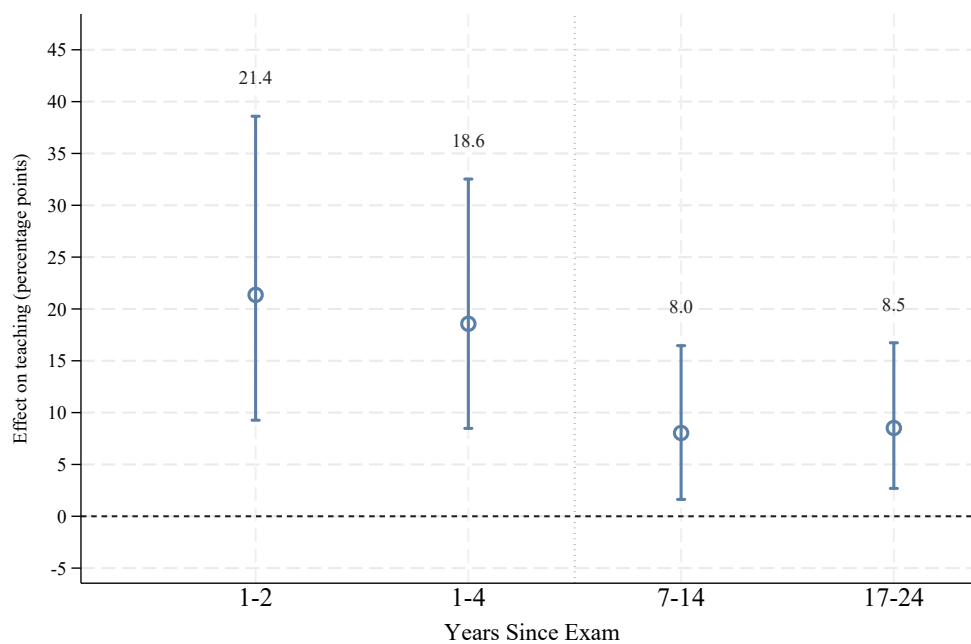
(c) Teaching in 1930



(d) Teaching in 1940

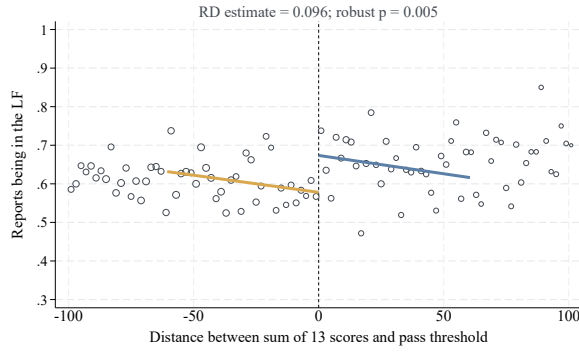
Notes: Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. Each panel reports the RD estimate and its robust p -value. The 1920 panel uses the 1916–1919 exam cohorts; the remaining panels use all 1916–1923 cohorts.

Figure 6: Immediate and Long-Run Effects on Teaching

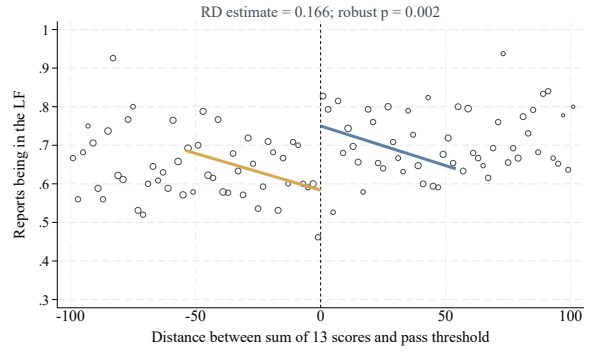


Notes: Markers show conventional point estimates, and vertical bars are 95 percent bias-corrected confidence intervals. Points 1 and 2 are measured in the 1920 census, point 3 in the 1930 census, and point 4 in the 1940 census. The one-to-two-year estimate uses the 1918–1919 cohorts, and the one-to-four-year estimate uses the 1916–1919 cohorts. The 1930 and 1940 estimates pool all 1916–1923 exam cohorts.

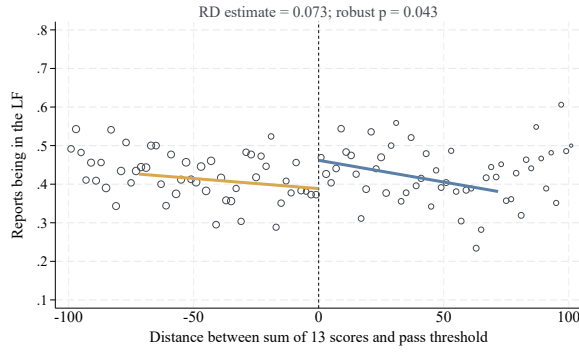
Figure 7: Effects on Labor Force Participation



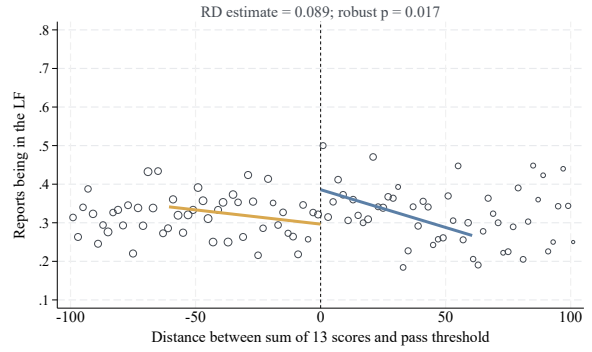
(a) Ever observed in the labor force by 1940



(b) Labor force participation in 1920



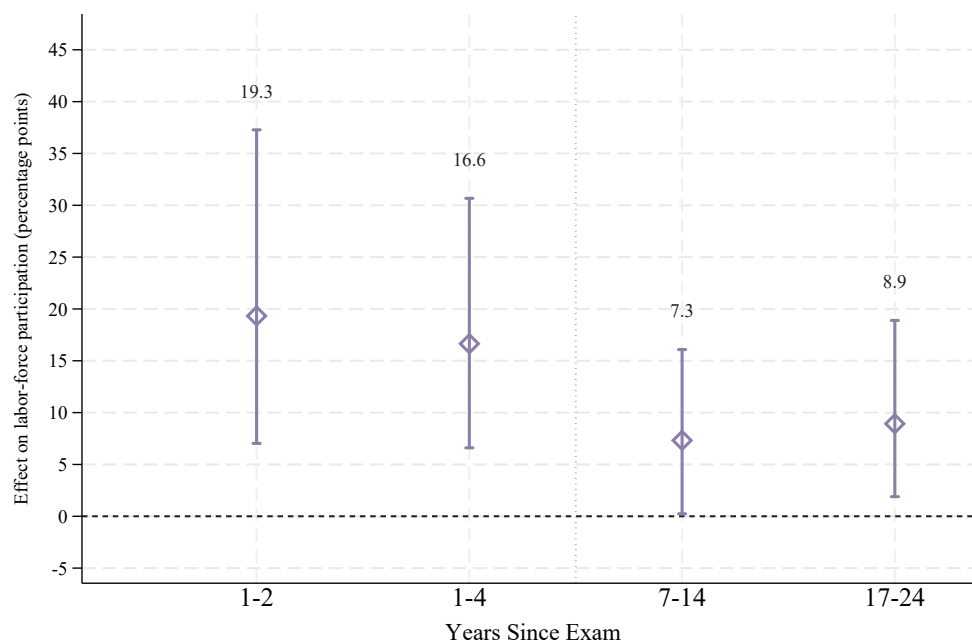
(c) Labor force participation in 1930



(d) Labor force participation in 1940

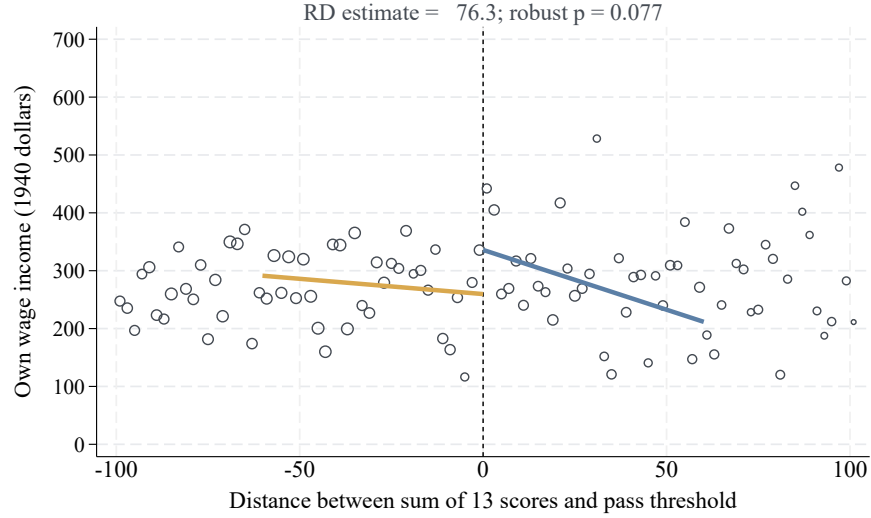
Notes: Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. Each panel reports the RD estimate and its robust p -value. The 1920 panel uses the 1916–1919 exam cohorts; the remaining panels use all 1916–1923 cohorts.

Figure 8: Immediate and Long-Run Effects on Labor Force Participation



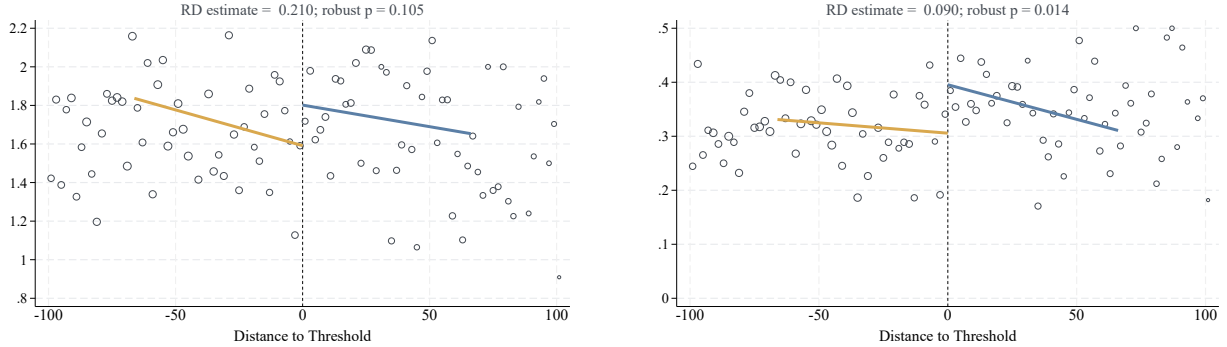
Notes: Markers show conventional point estimates, and vertical bars are 95 percent bias-corrected confidence intervals. Points 1 and 2 are measured in the 1920 census, point 3 in the 1930 census, and point 4 in the 1940 census. The one-to-two-year estimate uses the 1918–1919 cohorts, and the one-to-four-year estimate uses the 1916–1919 cohorts. The 1930 and 1940 estimates pool all 1916–1923 exam cohorts.

Figure 9: Effect on Own Wage Income in 1940



Notes: Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. The figure reports the RD estimate and its robust p -value.

Figure 10: Post-Exam Education

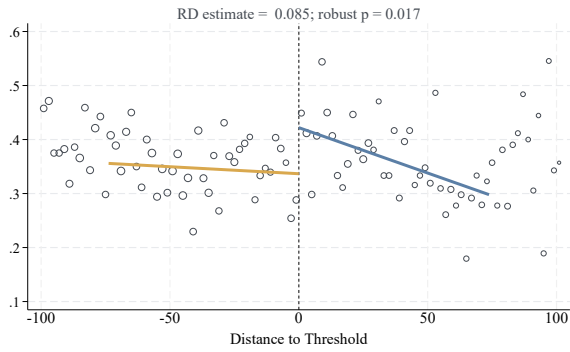


(a) Years of new education since exam

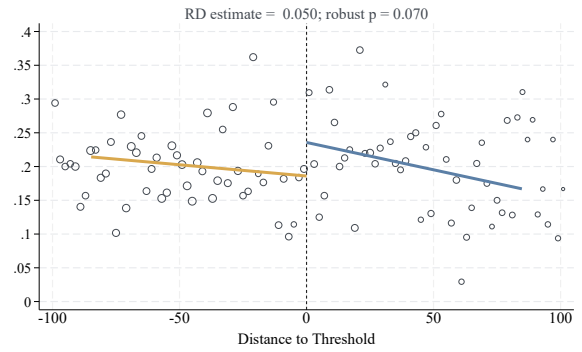
(b) Has 14+ years of schooling

Notes: Both panels are restricted to women whose reported education is no more than one year below the exam baseline. In panel (a) the outcome is completed years of education reported in 1940 minus years recorded at the exam; in panel (b) the outcome is an indicator for reaching at least fourteen years of completed schooling. Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. Each panel reports the RD estimate and its robust p -value.

Figure 11: Marriage



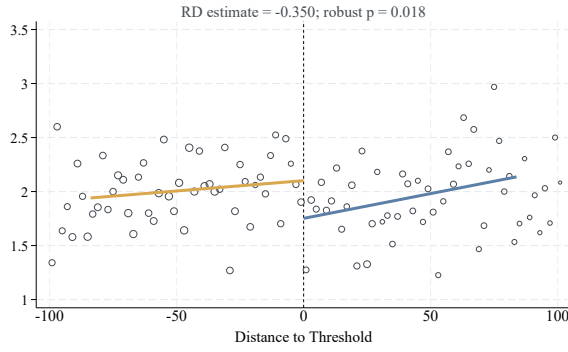
(a) Never married, 1930



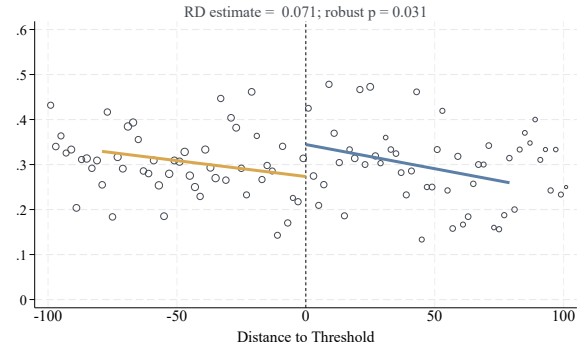
(b) Never married, 1940

Notes: Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. Each panel reports the RD estimate and its robust p -value.

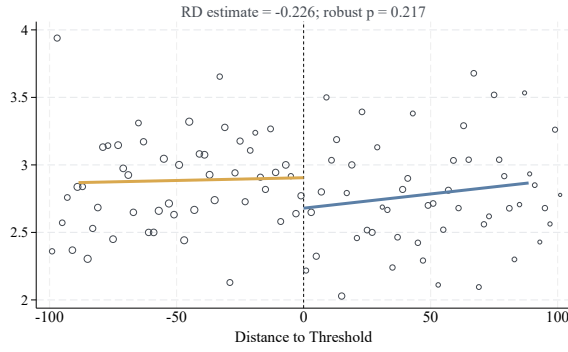
Figure 12: Fertility



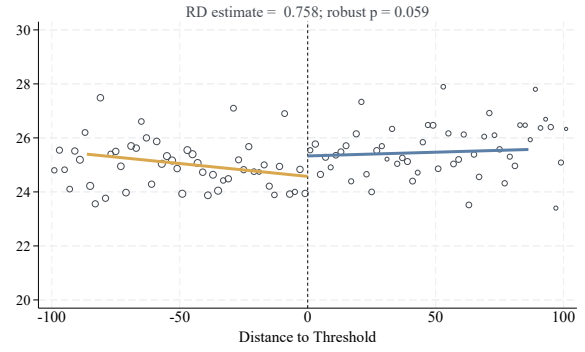
(a) Estimated fertility



(b) No children by 1940



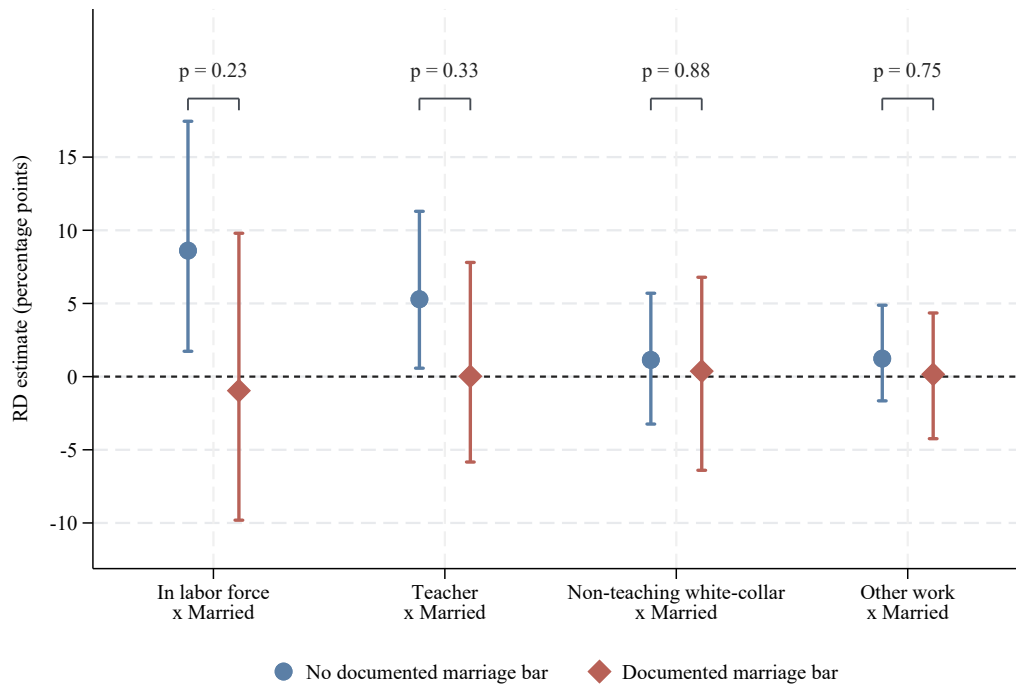
(c) Fertility conditional on positive



(d) Age at first birth

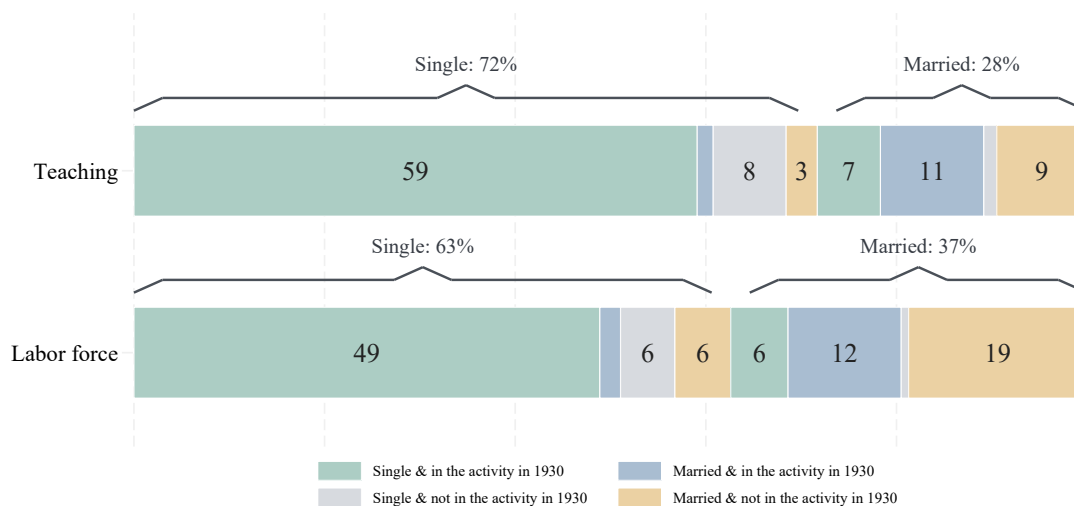
Notes: Every panel is restricted to women linked to both the 1930 and 1940 censuses. Hollow circles are outcome means in two-point bins, with marker area proportional to the number of observations. Linear fits use the MSE-optimal bandwidth, and bins are shown within 100 points. Each panel reports the RD estimate and its robust p -value.

Figure 13: Employment Effects by Current-Residence Marriage-Bar Status



Notes: Markers show conventional point estimates, and vertical bars are 95 percent bias-corrected confidence intervals. Each outcome equals one if a woman is observed working in the indicated occupation while married with her spouse present, in either 1930 or 1940. Non-teaching white-collar work includes occupation codes 1–99, excluding teaching, and 200–490; other work is non-teaching, non-white-collar employment. The bracket above each pair shows the p -value for the difference between the two estimates (bar minus no-bar), following [Calonico et al. \(2025\)](#).

Figure 14: Composition of 1940 teachers and labor-force participants, by 1930 status



Notes: Each bar describes the composition of women observed in the indicated activity (teaching; any labor-force participation) in the 1940 census, among applicant women linked to both the 1930 and 1940 censuses. Braces split each bar by 1940 marital status. Within each block, segments report the woman's 1930 status: single or married, crossed with whether she was observed in the activity in 1930. Values are percent of all 1940 participants in the activity; segments below 3 percent are unlabeled. Married includes spouse present and spouse absent.

13 Tables

Table 1: Female Teacher Applicants Compared to Other Groups in Illinois

Variable	Mean					Difference from teacher applicants			
	Teacher applicants (1)	IL general (2)	IL excl. Cook (3)	Eligible women (4)	Normal School (5)	– IL general (6)	– IL excl. Cook (7)	– Eligible (8)	– Normal School (9)
Age	13.22	13.21	13.21	13.34	13.23	4.86*** (0.04)	4.77*** (0.04)	1.85*** (0.04)	0.64*** (0.11)
Nonwhite (%)	0.4	1.6	1.8	1.6	1.3	-1.09*** (0.05)	-1.31*** (0.05)	-1.15*** (0.15)	-0.74*** (0.26)
Urban (%)	19.9	59.0	33.2	41.1	29.5	-37.00*** (0.30)	-11.45*** (0.30)	-19.27*** (0.47)	-9.40*** (1.19)
Foreign-born (%)	0.5	6.4	2.3	1.1	0.3	-4.17*** (0.06)	-1.43*** (0.06)	-0.76*** (0.11)	0.22 (0.15)
Number of siblings	3.19	3.37	3.46	3.14	3.03	-0.19*** (0.02)	-0.31*** (0.02)	0.15*** (0.03)	0.27*** (0.07)
Mother present (%)	92.9	86.1	85.2	92.6	92.2	-1.05*** (0.19)	-0.34* (0.19)	-2.34*** (0.29)	-0.20 (0.64)
Father present (%)	89.4	81.0	81.4	87.7	89.1	-1.02*** (0.23)	-0.90*** (0.23)	-1.89*** (0.34)	-0.75 (0.76)
Head in labor force (%)	93.0	92.2	92.8	90.9	90.8	-2.03*** (0.19)	-2.11*** (0.19)	0.37 (0.28)	1.96*** (0.71)
Head professional/technical (%)	2.7	2.6	2.2	4.5	6.2	0.29** (0.12)	0.69*** (0.12)	-1.49*** (0.19)	-3.53*** (0.68)
Head occupation score	18.30	20.53	18.30	20.51	19.89	-2.72*** (0.09)	-0.37*** (0.09)	-2.61*** (0.14)	-1.72*** (0.39)
Head education score	9.67	10.17	8.96	13.27	14.76	0.04 (0.12)	1.19*** (0.12)	-2.91*** (0.19)	-5.11*** (0.63)
Head is farmer (%)	49.3	23.9	39.0	28.5	40.0	24.11*** (0.37)	8.98*** (0.38)	18.39*** (0.54)	9.10*** (1.29)
In school (%)	90.0	74.5	78.3	91.4	89.3	10.84*** (0.29)	9.91*** (0.28)	-7.30*** (0.29)	0.01 (0.80)
Observations	18,691	1,445,186	829,431	71,826	2,390				

Notes: Childhood-household characteristics measured in the 1910 census (1920 as a fallback where no 1910 link exists), for Illinois teacher-exam applicants, Normal School applicants, and general IL women age ≤ 25 . “IL excl. Cook” drops Cook County; “Eligible” is women who could sit the exam—1910 IL girls (excl. Cook, ages 8–18) who link to 1940 with ≥ 10 years of schooling—weighted by the inverse 1910–1940 match probability to restore representativeness of the 1910 population. Comparison columns are reweighted to the applicants’ baseline-age distribution. Number of siblings is co-resident children minus one; head variables refer to the household head, with “professional/technical” occupation codes 1–99 and education score EDSCOR50. Difference columns are coefficients on a teacher-applicant indicator from a weighted regression with birth-year fixed effects; robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: RD Balance: Pre-Determined Covariates

Variable	Left Mean (1)	Jump (2)	p -value (3)
Age at exam	18.250	+0.047	0.685
Years of education at exam	11.256	+0.026	0.838
Urban (pre-exam residence)	0.214	−0.027	0.220
Mother present in household	0.927	+0.026*	0.087
Father present in household	0.904	+0.034*	0.073
Mother in labor force	0.083	−0.011	0.562
Father in labor force	0.968	−0.006	0.673
Any working woman in HH	0.241	−0.008	0.768
Number of siblings	3.013	−0.108	0.390
Household-head occ. score	19.006	−0.660	0.537
Household head is a farmer	0.475	+0.023	0.639
Birth year	1902.0	+0.173	0.287
Exam year	1920.0	+0.193	0.154
Matched to 1920 census (exam $\leq$ 1919)	0.852	+0.030	0.304
Matched to 1930 census	0.654	+0.027	0.365
Matched to 1940 census	0.597	−0.029	0.340
Matched to any subsequent census	0.784	+0.013	0.700

Notes: Each row is a separate local-linear RD regression of the listed pre-determined covariate on the recentered running variable, using the MSE-optimal bandwidth. The final four rows test whether passing affects linkage to the 1920, 1930, or 1940 census or to any subsequent census. The 1920 row is restricted to exam cohorts through 1919; the pooled measure captures linkage to the 1920, 1930, or 1940 census for exam cohorts through 1919 and linkage to the 1930 or 1940 census for exam cohorts after 1919. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effects on Teaching

	Ever a Teacher (1)	1920 Census 1918–1919 (2)	1920 Census 1916–1919 (3)	1930 Census (4)	1940 Census (5)
RD Estimate	+0.136***	+0.214***	+0.186***	+0.080**	+0.085***
<i>p</i> -value	< 0.001	0.001	< 0.001	0.017	0.007
Left Mean	0.382	0.471	0.454	0.240	0.141
Eff. <i>N</i>	4,174	1,105	1,710	3,583	3,012
<i>N</i>	12,492	2,609	5,084	10,494	9,368

Notes: Each column reports the RD estimate of the effect of passing the certification exam on the probability of being observed as a teacher, by local linear regression around the recentered cumulative score in the main specification with a triangular kernel and the MSE-optimal bandwidth (Calonico et al., 2017); bias-corrected *p*-values. “Ever a Teacher” equals one if a woman is observed teaching in any available census through 1940; the 1920 measures are available only for the early cohorts. Column 2 restricts to the 1918–1919 exam cohorts, observed one to two years after the exam; Column 3 uses all 1916–1919 cohorts observed in 1920. “Left Mean” is the outcome mean just below the cutoff; “Eff. *N*” is the number of observations within the bandwidth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects on Labor Force Participation

	Ever in LF (1)	1920 Census 1918–1919 (2)	1920 Census 1916–1919 (3)	1930 Census (4)	1940 Census (5)	Own Income 1940 (\$) (6)
RD Estimate	+0.096***	+0.193***	+0.166***	+0.073**	+0.089**	+76.3*
<i>p</i> -value	0.005	0.004	0.002	0.043	0.017	0.077
Left Mean	0.590	0.567	0.610	0.400	0.302	273.3
Eff. <i>N</i>	4,102	998	1,509	4,026	3,112	3,086
<i>N</i>	12,492	2,609	5,084	10,494	9,368	9,271

Notes: See notes to Table 3. “Ever in LF” equals one if a woman is observed in the labor force in any available census through 1940; the 1920 measures are available only for the early cohorts. Column 2 restricts to the 1918–1919 exam cohorts, observed one to two years after the exam; Column 3 uses all 1916–1919 cohorts observed in 1920. The next two outcomes are indicators for being in the labor force in the relevant census. The final column reports the woman’s own wage income in 1940.

Table 5: Marriage

	Never Married 1930 (1)	Never Married 1940 (2)	Married, Spouse Present, 1940 (3)	Divorced 1940 (4)	Widowed 1940 (5)
RD Estimate	+0.085**	+0.050*	−0.063**	+0.005	+0.004
<i>p</i> -value	0.017	0.070	0.035	0.557	0.532
Left Mean	0.348	0.193	0.755	0.011	0.026
Eff. <i>N</i>	4,125	4,271	3,808	4,416	3,612
<i>N</i>	10,494	9,368	9,368	9,368	9,368

Notes: See notes to Table 3. Outcomes are indicators for the listed marital status in the relevant census. Married, spouse present corresponds to marital-status code 1.

Table 6: Effects on Fertility

	Estimated Fertility (1)	No Children by 1940 (2)	Fertility Conditional on Positive (3)	Age at First Birth (4)
RD Estimate	−0.350**	+0.071**	−0.226	+0.758*
<i>p</i> -value	0.018	0.031	0.217	0.059
Left Mean	2.073	0.284	2.893	24.748
Eff. <i>N</i>	3,788	3,617	2,794	2,671
<i>N</i>	8,464	8,464	5,961	5,787

Notes: See notes to Table 3. Every column is restricted to women linked to both the 1930 and 1940 censuses. Estimated fertility is the number of own children observed across the linked census panel; no children equals one when estimated fertility is zero. The conditional-fertility column is restricted to women with positive estimated fertility. Age at first birth is the earliest inferred first-birth age across the two census waves.

Table 7: Women’s Earnings and Household Wage Income

	Household Wage Income Accounting			Women’s Wage Income by Spouse Status	
	Household Total (1)	Woman’s Own Wage Income (2)	Husband-Contributed Wage Income (3)	Spouse Present (4)	No Spouse Present (5)
RD Estimate	+80.0	+64.2	+8.3	+12.8	+51.4
<i>p</i> -value	0.166	0.102	0.613	0.302	0.241
Mean below cutoff	1,095.2	330.8	757.1	69.5	261.3
Effective <i>N</i>	3,659	3,659	3,659	3,659	3,659
<i>N</i>	7,506	7,506	7,506	7,506	7,506

Notes: See notes to Table 3. Wage income is measured in the 1940 census. Household wage income equals the woman’s own wage income plus her co-resident husband’s wage income. Women’s own-wage income is defined for all women regardless of marital status. Appendix Table A4 decomposes husband-contributed income into spouse-presence and conditional-income margins. Each outcome presented uses the optimal bandwidth selected for the household-income outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Persistent Employment and Occupation, Overall and by Marital Status

	All 1930–1940 (1)	Ever No Spouse 1930–1940 (2)	Ever Spouse Present 1930–1940 (3)
<i>Panel A: Labor Force Participation</i>			
RD Estimate	+0.089**	+0.062*	+0.050*
<i>p</i> -value	0.019	0.066	0.058
Left Mean	0.469	0.362	0.122
Eff. <i>N</i>	3,738	4,201	3,534
<i>N</i>	11,374	11,374	11,374
<i>Panel B: Teaching</i>			
RD Estimate	+0.111***	+0.088***	+0.033*
<i>p</i> -value	0.002	0.009	0.053
Left Mean	0.254	0.211	0.049
Eff. <i>N</i>	3,534	3,797	3,534
<i>N</i>	11,374	11,374	11,374
<i>Panel C: Other Non-Teaching White-Collar Work</i>			
RD Estimate	−0.000	−0.031	+0.018
<i>p</i> -value	0.925	0.118	0.364
Left Mean	0.169	0.127	0.050
Eff. <i>N</i>	6,544	4,680	4,735
<i>N</i>	11,374	11,374	11,374
<i>Panel D: Other Non-Teaching, Non-White-Collar Work</i>			
RD Estimate	+0.008	+0.016	−0.001
<i>p</i> -value	0.576	0.139	0.978
Left Mean	0.061	0.032	0.028
Eff. <i>N</i>	6,347	5,372	5,417
<i>N</i>	11,374	11,374	11,374

Notes: See notes to Table 3. Column 1 equals one if a woman is observed in the listed occupation in either the 1930 or 1940 census, regardless of spouse status. Columns 2 and 3 equal one if she is observed in the listed occupation and spouse-status cell in either census. Columns are not mutually exclusive since women’s marital status may change between censuses. “Spouse Present” requires a co-resident spouse; “No Spouse” includes never-married, divorced, widowed, and married women whose spouse is not recorded in the household. Other non-teaching white-collar work includes occupation codes 1–99 (excluding teachers, code 93) and 200–490. Other non-teaching, non-white-collar work is the remaining labor-force participation outside teaching and those white-collar occupations. Appendix Table A5 reports the corresponding wave-specific estimates.

Table 9: Post-Exam Schooling and Long-Run Employment Persistence

	New Education Since Exam (1)	14+ Years Schooling (2)	New Education $\times$ Not in LF 1930–1940 (3)	New Education $\times$ In LF, Not a Teacher, 1930–1940 (4)	New Education $\times$ Teacher 1930–1940 (5)
RD Estimate	+0.210	+0.090**	−0.006	−0.056	+0.253**
<i>p</i> -value	0.105	0.014	0.925	0.375	0.035
Left Mean	1.651	0.308	0.690	0.317	0.657
Eff. <i>N</i>	3,124	3,124	3,227	3,424	2,913
<i>N</i>	8,581	8,581	8,581	8,581	8,581

Notes: See notes to Table 3. New education is 1940 completed years of schooling minus education recorded at the exam, reported among women whose 1940 schooling is no more than one year below the education recorded at the exam. “14+ Years Schooling” equals one for women with at least 14 completed years of schooling in the same restricted education sample. Columns 3–5 partition interact new education with 3 forms of labor force participation: not observed in the labor force in either census; observed in the labor force in at least one census but not as a teacher in either; and observed as a teacher in at least one census.

A Appendix Tables

Table A1: Income-Data Availability at the Threshold

	Husband Income Missing Unconditional	Husband Income Missing Spouse Present	Household Income Observed
RD Estimate	+0.004	+0.015	+0.010
<i>p</i> -value	0.996	0.659	0.611
Left Mean	0.082	0.108	0.804
Eff. <i>N</i>	4,063	3,114	3,715
<i>N</i>	9,271	6,932	9,368

Notes: Each column reports a local-linear RD estimate at the certification threshold using a triangular kernel and an outcome-specific MSE-optimal bandwidth. Column 1 indicates that a woman has a co-resident husband whose wage income is unavailable, among women with valid own wage income. Column 2 reports the probability that husbands' income is missing, conditional on having a co-resident husband and a valid own-income (0 or positive). "Eff. *N*" is the number of observations used within the optimally selected estimation bandwidth.

Table A2: Husband Employment and Wage Income, Conditional on Spouse Presence

Outcome	RD Estimate	<i>p</i> -value	Mean Below Cutoff	Eff. <i>N</i>	<i>N</i>
Husband in labor force	−0.023**	0.022	0.989	3,294	6,986
Husband is a farmer	−0.044	0.197	0.286	2,218	6,986
Husband has positive wage income	+0.017	0.595	0.647	3,450	6,237
Husband wage income	+117.5	0.259	1,095.6	3,237	6,237
Husband wage income, conditional on positive	+122.9	0.346	1,692.5	1,945	4,098

Notes: Each row reports a local-linear RD estimate conditional on having a linked co-resident husband and observing the relevant outcome. "Farmer" includes farmers and farm managers (1950 occupation codes 100 and 123). Husband wage income includes valid zeros; the final row is additionally restricted to positive wage income. "Eff. *N*" is the number of observations within the outcome-specific MSE-optimal bandwidth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Husband Characteristics and Intergenerational Income Rank

Outcome	RD Estimate	<i>p</i> -value	Mean Below Cutoff	Eff. <i>N</i>	<i>N</i>
Husband's years of education	+0.205	0.494	10.468	3,336	6,923
Husband has 14+ years of education	+0.010	0.762	0.172	3,807	6,923
Husband's occupation score	+0.456	0.567	26.221	2,913	6,858
Husband is a teacher	+0.031**	0.046	0.030	2,887	6,986
Age gap: husband minus wife	+0.280	0.384	2.561	3,552	6,986
Father's income-score rank	−2.418	0.196	45.493	6,317	11,669
Husband's wage-income rank	+2.106	0.539	47.853	3,067	6,234
Husband rank minus father rank	+4.018	0.300	5.584	2,406	4,741
Husband rank exceeds father rank	+0.053	0.275	0.503	2,266	4,741

Notes: Each row reports a local-linear RD estimate using an outcome-specific MSE-optimal bandwidth. Husband characteristics are measured in the 1940 census and are conditional on a linked co-resident husband and observation of the relevant characteristic. Father's income-score rank uses his positive IPUMS occupation score in the woman's pre-exam census and ranks him against White fathers in the full-count 1910 or 1920 census who had a child born in the same year as the woman. Husband's wage-income rank uses actual 1940 wage income and ranks him against White married, spouse-present men born in the same year in the full-count 1940 census; valid zero income is retained. Rank differences require both ranks to be observed. "Eff. *N*" is the number of observations within the outcome-specific bandwidth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Decomposition of Husband-Contributed Wage Income

Component	Change in Underlying Margin	Contribution to RD Estimate (dollars)
Total husband-contributed wage income		+8.3
Change in spouse presence	−0.078	−81.3
Change in conditional husband wage income	+138.2	+100.4
Interaction		−10.7

Notes: The table decomposes the RD estimate for husband-contributed wage income into an extensive spouse-presence margin, an intensive conditional-husband-income margin, and their interaction. The decomposition uses common-bandwidth triangular-kernel local-linear cutoff means on the household-income sample and sums to the total up to rounding: $-81.3 + 100.4 - 10.7 = 8.3$.

Table A5: Employment and Occupation, by Marital Status and Census Wave

	No Spouse 1920 (1)	Spouse Present 1920 (2)	No Spouse 1930 (3)	Spouse Present 1930 (4)	No Spouse 1940 (5)	Spouse Present 1940 (6)
<i>Panel A: Labor Force Participation $\times$ Marital Status</i>						
RD Estimate	+0.174***	−0.000	+0.052	+0.021	+0.046*	+0.043*
<i>p</i> -value	0.002	0.993	0.108	0.280	0.088	0.084
Left Mean	0.604	0.006	0.333	0.067	0.208	0.094
Eff. <i>N</i>	1,436	2,280	4,125	4,712	4,232	3,204
<i>N</i>	5,084	5,084	10,494	10,494	9,368	9,368
<i>Panel B: Teaching $\times$ Marital Status</i>						
RD Estimate	+0.181***	+0.006	+0.081**	−0.003	+0.049**	+0.038**
<i>p</i> -value	0.001	0.205	0.013	0.756	0.049	0.028
Left Mean	0.451	0.003	0.204	0.036	0.107	0.034
Eff. <i>N</i>	1,710	2,115	3,583	5,296	3,504	3,112
<i>N</i>	5,084	5,084	10,494	10,494	9,368	9,368
<i>Panel C: Other Non-Teaching White-Collar Work $\times$ Marital Status</i>						
RD Estimate	−0.013	+0.000	−0.043**	+0.021*	−0.018	+0.000
<i>p</i> -value	0.717	0.595	0.028	0.086	0.272	0.965
Left Mean	0.110	0.000	0.114	0.021	0.078	0.042
Eff. <i>N</i>	2,066	942	4,282	4,441	3,948	4,669
<i>N</i>	5,084	5,084	10,494	10,494	9,368	9,368
<i>Panel D: Other Non-Teaching, Non-White-Collar Work $\times$ Marital Status</i>						
RD Estimate	−0.008	−0.005	+0.010	+0.005	+0.012	+0.004
<i>p</i> -value	0.738	0.219	0.314	0.463	0.175	0.559
Left Mean	0.046	0.003	0.016	0.010	0.023	0.023
Eff. <i>N</i>	2,336	3,436	5,243	6,244	4,669	3,612
<i>N</i>	5,084	5,084	10,494	10,494	9,368	9,368

Notes: See notes to Table 8. Columns report separate outcomes for the 1920, 1930, and 1940 censuses. The 1920 columns are restricted to the 1916–1919 exam cohorts.

A.1 Heterogeneity by Farm vs. Non-Farm Household

Table A6: Heterogeneity by Farm vs. Non-Farm Household

Outcome	Non-Farm		Farm		Diff. p
	Estimate	N	Estimate	N	
Teacher, 1930	+0.083**	4,836	+0.101**	5,157	0.86
Teacher, 1940	+0.089**	4,285	+0.097**	4,641	0.93
In labor force, 1930	+0.043	4,836	+0.110**	5,157	0.50
In labor force, 1940	+0.068	4,285	+0.105**	4,641	0.72
Years of schooling	+0.131	4,280	+0.691***	4,636	0.23
Ever married, 1930	−0.059	4,836	−0.123**	5,157	0.49
Ever married, 1940	−0.067	4,285	−0.036	4,641	0.71
Est. fertility	−0.170	3,850	−0.507**	4,226	0.40
Est. fertility positive	+0.025	2,625	−0.364	3,093	0.37
Age at first birth	+0.795	2,545	+0.826*	3,004	0.98
No children (1930–40)	+0.065	3,850	+0.085**	4,226	0.84
Own wage income	−10.5	4,240	+168.3***	4,591	0.19

Notes: Subgroup RD estimates (MSE-optimal bandwidths, HC3 standard errors), splitting the sample by whether the woman’s father was a farmer (occupation code 100) or held any other occupation. Father’s occupation is taken from the woman’s 1910 household—recovered via her own childhood census record where her baseline census link falls later—covering roughly 93 percent of the sample; the split is close to even (49.7 percent farm). “Diff. p ” is the p -value for the difference between the two subgroup effects, following [Calonico et al. \(2025\)](#). N is per subgroup. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.2 Heterogeneity by Marriage-Bar County

Table A7: Heterogeneity by Marriage-Bar County

Outcome	No Bar		Bar County		Diff. p
	Estimate	N	Estimate	N	
Teacher, 1930	+0.088*	5,498	+0.110**	3,559	0.83
Teacher, 1940	+0.125***	4,698	+0.056	3,075	0.48
In labor force, 1930	+0.060	5,498	+0.120**	3,559	0.59
In labor force, 1940	+0.078	4,698	+0.082	3,075	0.97
Years of schooling	+0.571**	4,693	+0.266	3,072	0.53
Ever married, 1930	−0.079	5,498	−0.151***	3,559	0.48
Ever married, 1940	−0.041	4,698	−0.092*	3,075	0.59
Est. fertility	−0.322	4,235	−0.323	3,280	0.99
Est. fertility positive	−0.365	2,947	−0.081	2,289	0.54
Age at first birth	+1.643***	2,877	+0.334	2,199	0.30
No children (1930–40)	+0.028	4,235	+0.112*	3,280	0.44
Own wage income	+47.2	4,639	+104.6*	3,048	0.64
Married $\times$ Teacher, 1940	+0.072**	4,698	−0.016	3,075	0.09
Unmarried $\times$ Teacher, 1940	+0.050	4,698	+0.064	3,075	0.87

Notes: Subgroup RD estimates (MSE-optimal bandwidths, HC3 standard errors). A county is a “bar county” if any city in it ever implemented or was reported operating a married-teacher bar. Counties are classified by the woman’s census county of residence: 1930 outcomes use 1930 residence, 1940 outcomes use 1940 residence, and the dual-census fertility and age-at-first-birth measures use any bar-county residence in 1930–1940; subgroups are defined for Illinois residents in the relevant census. The married- and unmarried-teacher rows split teaching in 1940 by spouse presence in 1940. “Diff. p ” is the p -value for the difference between the two subgroup effects, following [Calonico et al. \(2025\)](#). N is per subgroup. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Equality of the Teaching and Labor-Force Effects

Table A8: Equality of the Teaching and Labor-Force Effects

Outcome	RD estimate (pp)	<i>p</i> -value
Ever observed teaching by 1940	+13.6	<0.001
Ever observed in the labor force by 1940	+9.6	0.005
Difference (teaching – labor force)	+3.8	0.131
95% confidence interval	[−1.3, 9.8]	

Notes: Conventional local-linear RD estimates (MSE-optimal bandwidth, triangular kernel, HC3 standard errors). “Ever observed teaching (in the labor force) by 1940” equals one if the woman is observed as a teacher (in the labor force) in any subsequent census. The difference row tests equality of the two effects: it is the RD estimate on the difference outcome (ever teaching minus ever in the labor force), estimated on the same women so that the standard error accounts for the covariance between the two outcomes; the reported *p*-value is the robust *p*-value for the null that the effects are equal.

A.4 Aggregate Implications Under Alternative Scalings

Table A9: Share of the Aggregate Change Explained by Additional Early-Career Entry

Additional early-career entry into	Growth in entry (pp)	Share of the aggregate change	
		Prime-age LFP	Completed fertility
Certification margin only, downstate IL (no scaling)	1.1	1%	0.2%
All teachers, U.S.	1.5	1–3%	0.3–0.8%
All teachers, rural U.S.	2.4	6–13%	0.5–1.3%
All professional occupations, U.S.	2.5	2–6%	0.5–1.3%
All white-collar work, U.S. (if clerical $\approx$ teaching)	16.4	16–36%	4–8%

Notes: Each share is (growth in entry) $\times$ (effect per entrant) $\div$ (aggregate change). Row 1 uses the reduced-form estimates (+8.9 pp on 1940 labor force participation; -0.35 children) and counts the 1.1 percent of the downstate 1897–1902 birth cohorts who sat the exam and passed within 3 percentage points of the threshold (0.113×0.099); its denominators are downstate-Illinois women (participation 10.5 to 21.5 percent at ages 35–44; completed fertility 4.83 to 2.67 children at ages 45–54). The remaining rows scale per certified woman, dividing the reduced form by the 0.50 first stage on certification (+17.8 pp; -0.70 children) for the lower bound and by the 0.214 first stage on teaching in the 1920 census (1918–1919 exam cohorts; Table 3, Column 2; +41.6 pp; -1.64 children) for the upper bound, and count the growth in entry between women aged 18–24 in 1880 and in 1920: teaching 2.3 to 3.8 percent (rural 3.0 to 5.4), professional occupations 2.7 to 5.2 percent, and all white-collar work 3.4 to 19.7 percent. Their denominators are national women aged 35–44 in 1900 and 1940 (participation 15.7 to 27.1 percent; rural 12.7 to 17.4) and aged 45–54 (completed fertility 4.71 to 2.76 children; rural 5.15 to 3.24). Every scaled row applies the per-career effect only to the 61 percent of new entrants who came from non-employment; the 39 percent who came from other gainful work were already working and are counted as contributing nothing. That origin split is measured for white-collar entry as a whole and applied to each row. Counts and denominators come from the full-count censuses.

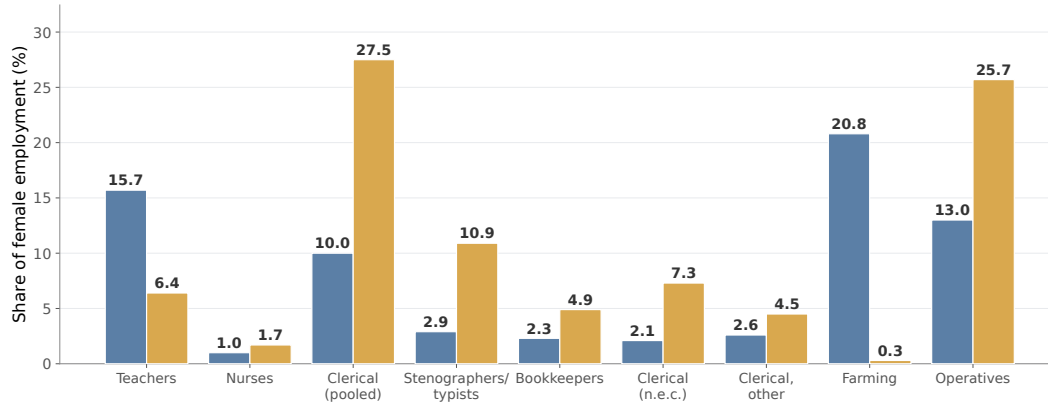
B Appendix Figures

Figure B1: Teaching pay versus state civil-service pay, 1922

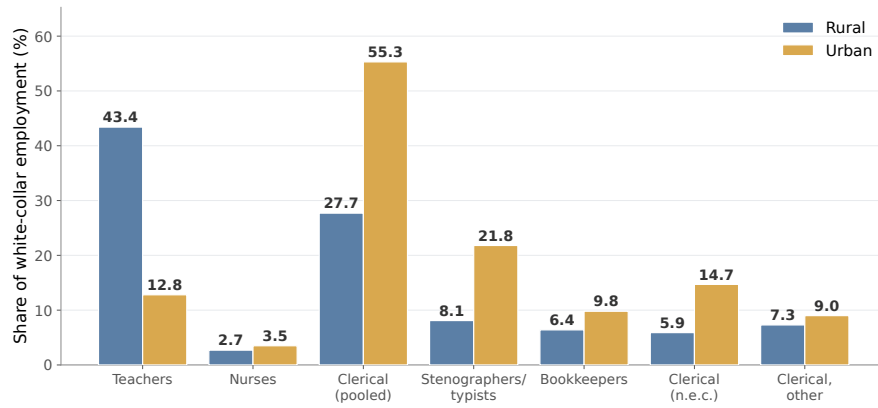


Notes: Bars show monthly salary ranges of women-relevant Illinois civil-service job classes, from the 1922 classification of the Illinois State Civil Service Commission, colored by the stated education requirement; (+ board) indicates the class included maintenance (board and lodging). Civil-service salaries are paid year-round. Vertical lines show the average teacher salary in 1922 (men and women pooled, as the civil-service classes report no sex breakdown) from the State Superintendent's biennial reports. Blue: Illinois excluding Cook County (\$1,046 per year; approximately eight-month term). Purple: Illinois including Cook County (\$1,343 per year; teacher-weighted term of roughly 8.5 months, as Chicago's term ran 9.5 months). For each average, the solid line divides annual salary by months in session and the dashed line spreads it over twelve months; civil-service salaries are year-round, so the two lines bracket the comparison.

Figure B2: Women's occupation shares by rural/urban residence, 1920



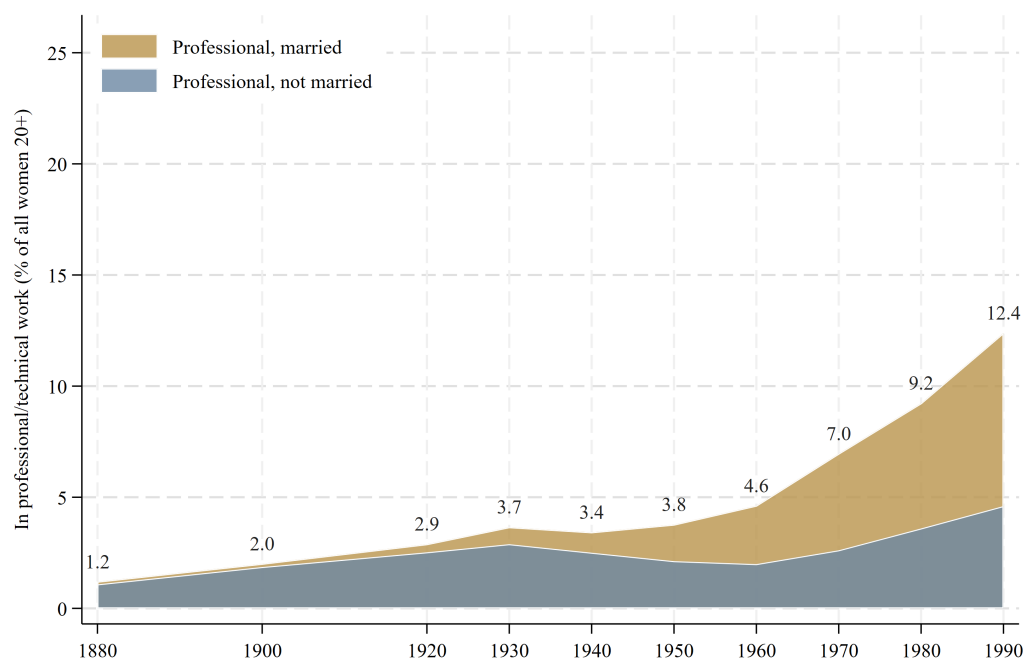
(a) Share of all female employment



(b) Share of female white-collar employment

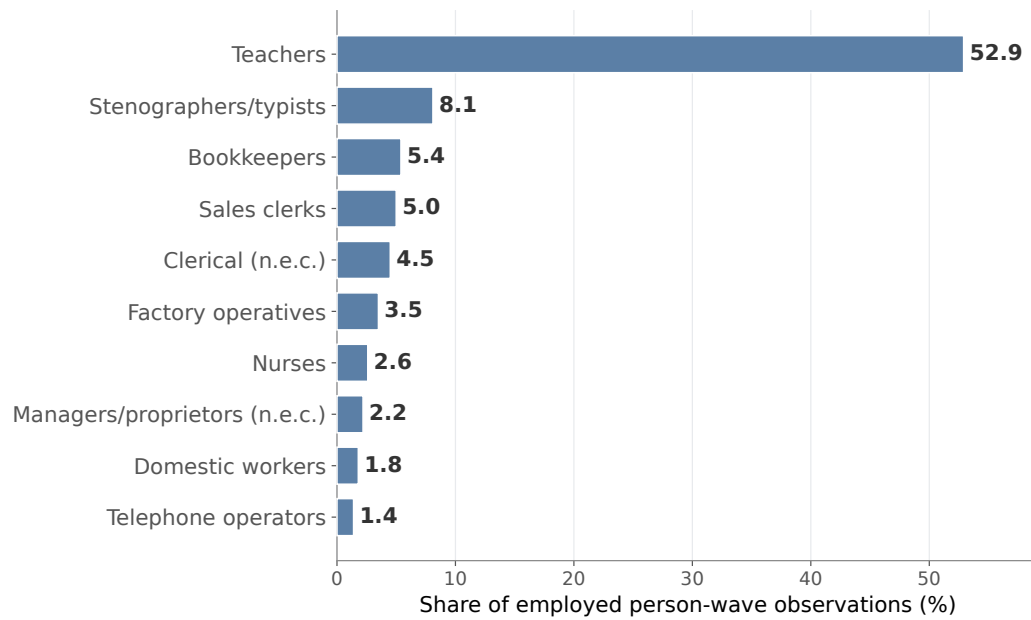
Notes: Full-count 1920 census: white women aged 19 and older in the labor force with a gainful occupation. Clerical (pooled) is occ1950 codes 300–390; the separated clerical bars (stenographers/typists, bookkeepers, clerical n.e.c., other) partition it. White collar (panel b) is occ1950 1–99 and 200–490 (professional, managerial, clerical, and sales occupations); farming and operative occupations are therefore shown only in panel (a).

Figure B3: The rise of professional and technical work among women, by marital status



Notes: The figure plots the share of all women aged 20 and older employed in professional and technical work (1950 occupation codes 1–99), stacked by marital status (currently married versus not currently married); value labels give the total. Data are the IPUMS 1% samples of the decennial censuses for 1880–1990 (the 1910 sample is excluded because its enumeration overstates women’s gainful employment); estimates are weighted by person weights.

Figure B4: Occupations of exam-sample women, pooled 1930–1940



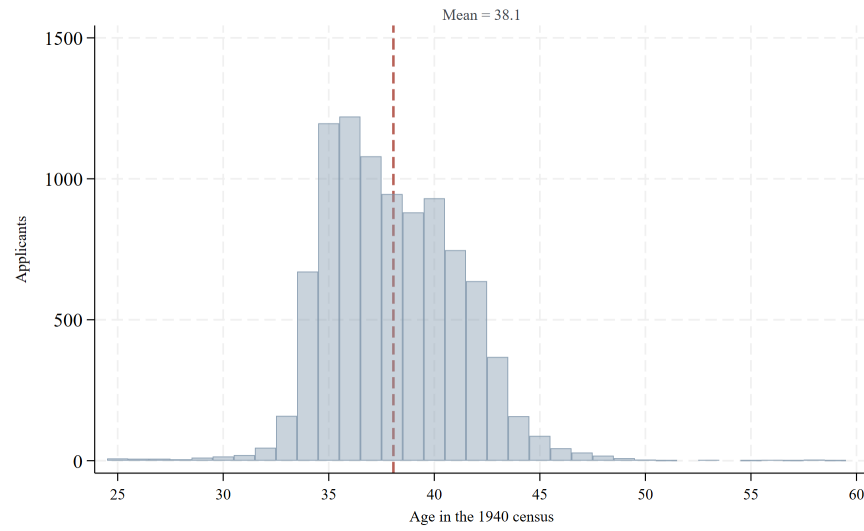
Notes: Top ten occupations among exam-sample women observed in the labor force with a gainful occupation, pooling person-wave observations across the 1930 and 1940 censuses. Shares are of all employed person-wave observations.

Figure B5: Example Examination Ledger Page

For Examination held <u>March 16 & 17, 1922.</u>						SECOND GRADE													
To the County Superintendent of <u>Adams</u> County:																			
We hereby certify that the following grades are the true and correct results of the examination for teachers' county certificates held in your county. Valid grades earned in the first preceding examination are recorded on the blue line in black; in the second preceding examination, on the blue line in red. You are hereby authorized to grant certificates to all applicants who have completed the examination with no grade below 60 and with an average of 75 if for second grade, or 80 if for other grade of certificate.																			
Chairman. Secretary.																			
Application Number	Name of Applicant	Address	Grade of Certificate applied for	Age of Applicant	Preparation: A. 2 years in recognized higher institution; B. 1 year in recognized higher institution; C. 4 years high school; D. 3 years high school; E. 2 years high school	Physiology	Pennmanship	Grammar	Reading	Orthography	Geography	United States History	Illinois History	Civics	State Course	Arithmetic	Pedagogy	General Science	Average
677	Reeder, Rena	Quincy	Second	18-9	E	75	85	80	83	78	70	80	80	45	75	81	89		
678	McQuaid, Sadie Leona	Augusta	Second	19-0	E	45	70	39	60	60	62	30	83	7	75	50	50	49	
679	Gamble, Donald E.	Barry	Second	18-9	D	60	75	62	50	60	55	16	80	5	50	62			
680	Crank, Neva Jeanette	Menden	Second	19-2	D	64	80	85	50	80	67	75	77	27	72	50	70	85	
681	Reece, Leora Blanche	Lorraine	Second	19-2	D	64	70	37	50	68	55	14	81	17	85	34	60	60	
682	Finney, Daisy May	Ursa	Second	18-10	E	77	75	90	75	88	64	22	87	27	86	100	77	55	
683	Poland, Marie	Clayton	Second	19-7	D	94	90	60	65	75	70	25	81	20	82	42	78	60	
684	Stevens, Harriett	Clayton	Second	17-10	C	70	85	65	60	78	72	47	78	25	80	67	85	65	
685	Wolf, Katherine	Quincy	Second	18-0	E	54	75	47	50	60	66	20	83	29	84	67	76	65	
686	Huegge, Lorine	Quincy	Second	17-11	D	64	85	40	60	86	67	39	84	35	80	84	82	49	
687	Stewart, Flora Helen	Plainville	Second	19-1	D	90	90	84	68	89	69	37	87	35	90	100	85	60	
688	Kaltenbach, Alma	Payson	Second	17-9	E	60	80	50	55	77	65	24	89	16	75	67	75	49	
689	McMutt, Jacinth	Payson	Second	17-7	E	87	95	47	75	83	69	22	83	21	70	84	78	60	
690	Constable, Everett L.	Payson	Second	20-2	D	67	70	44	60	80	70	54	81	17	82	67	88	45	
691	Long, Marie Naomi	Payson	Second	19-6	D	77	70	62	50	78	66	52	84	29	84	75	86	50	

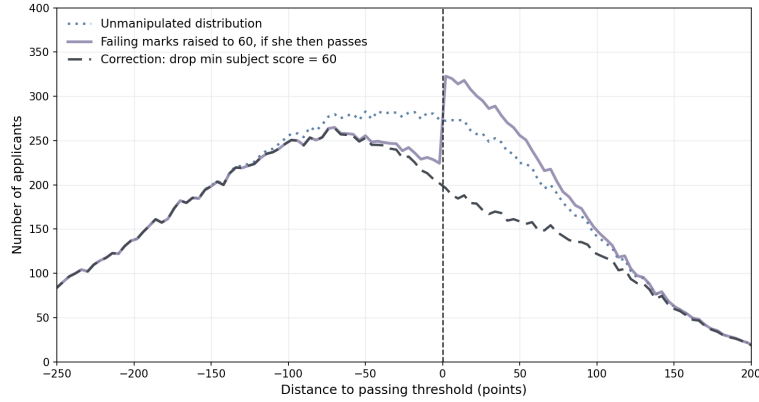
Notes: A page from the Report of the Illinois State Examining Board (Adams County, examination held March 1922). Each row records an applicant's application number, name, address, grade of certificate applied for, age, preparation code, and score on each of the thirteen subject examinations.

Figure B6: Age Distribution of Applicants in the 1940 Census

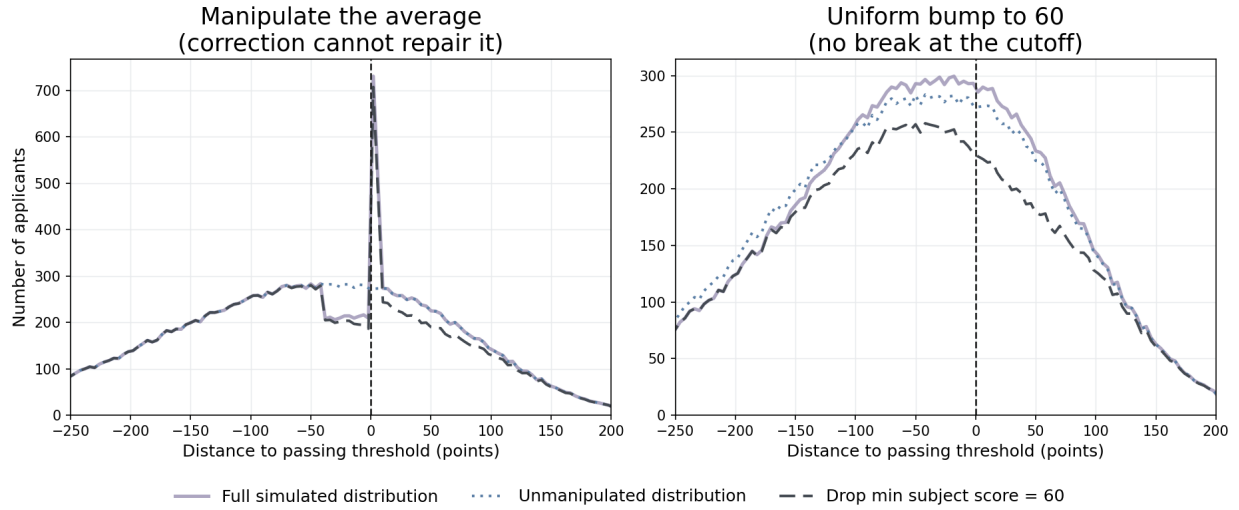


Notes: Distribution of applicants' ages in the 1940 census, among sample women linked to 1940. The dashed line marks the mean (38.1).

Figure B7: Simulated Grading Behavior



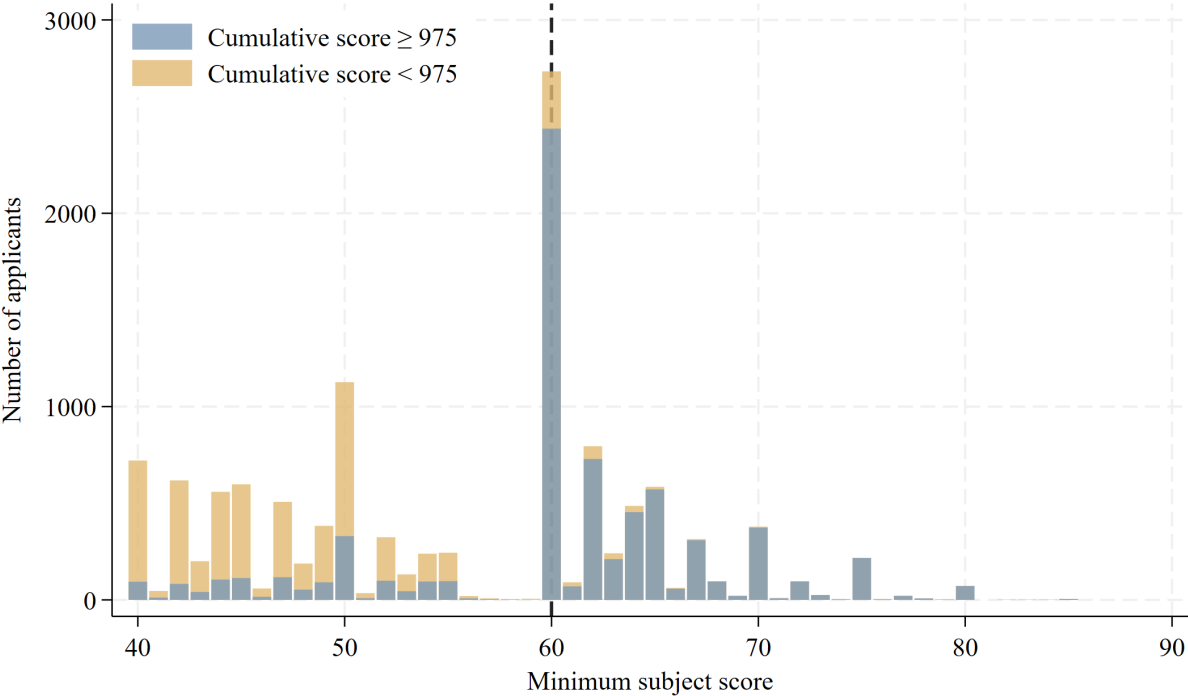
(a) Minimum bumped to 60 when this lets the candidate pass



(b) Manipulating the average (left) and a uniform bump to 60 (right)

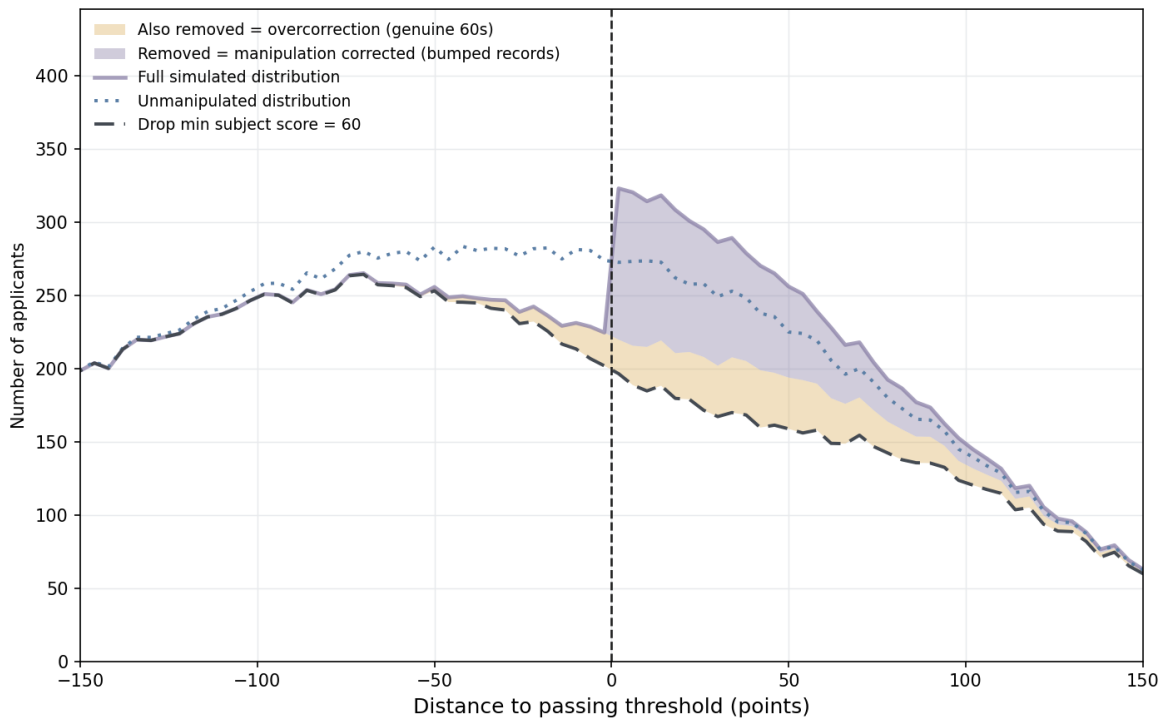
Notes: Each panel plots a simulated distribution of the cumulative exam score (the sum of the thirteen subject scores), recentered so that the passing threshold falls at a distance of zero, under a different grading regime. Each panel shows the full distribution (solid), the distribution after dropping candidates whose minimum subject score equals exactly 60 (dashed), and the true unmanipulated distribution (dotted). The regimes are: (a) a subject-60 bump applied only to candidates for whom this change would allow them to pass the exam; (b, left) direct manipulation of the average across the cutoff; and (b, right) a subject-60 bump applied to a share of candidates regardless of their average. Dropping minimum-score 60s recovers a smooth density under regime (a) and leaves the already-smooth density of the uniform bump smooth, but does not repair direct manipulation of the average. Because some candidates genuinely score at least 60, the exclusion rule also removes genuine, unmanipulated records, resulting in additional depletion of the recovered unmanipulated distribution (dotted lines); for more information, see Figure B9.

Figure B8: Minimum Subject Score and Grade Bumping



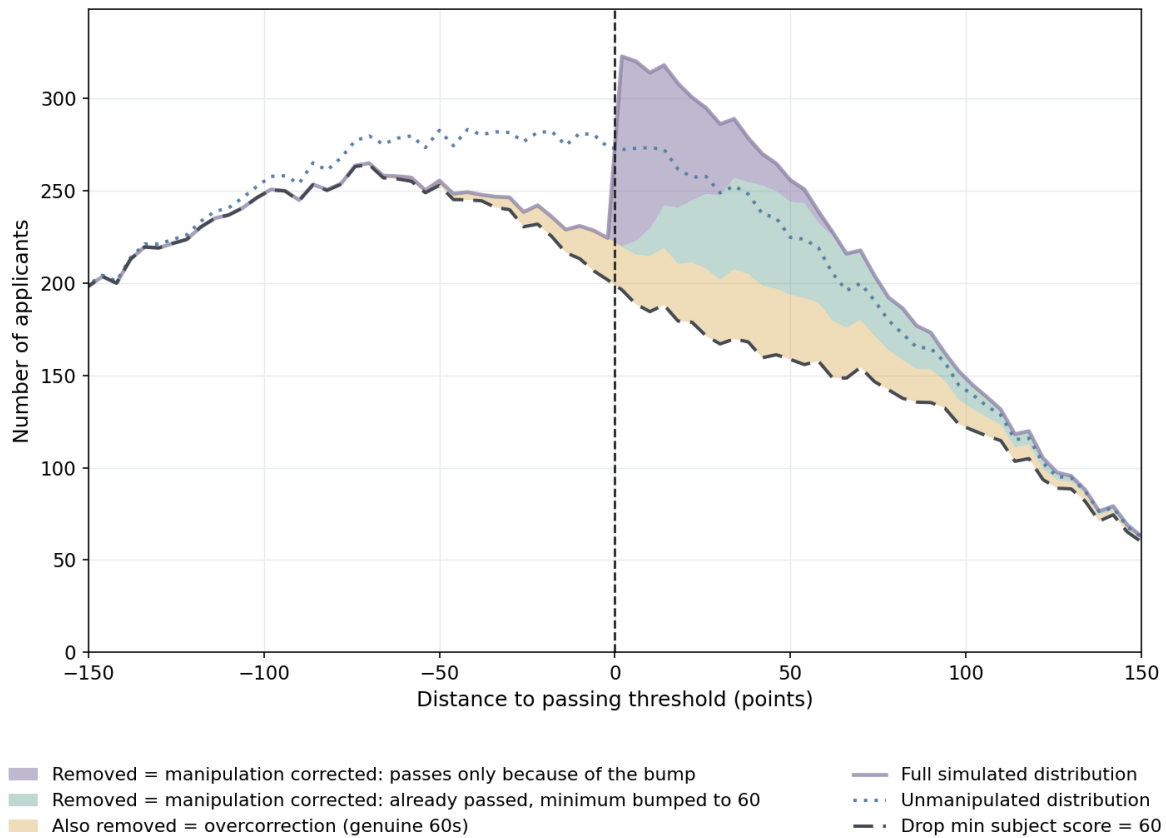
Notes: Distribution of the minimum subject score among first-attempt applicants with complete thirteen-subject exams, split by whether the candidate cleared the cumulative passing margin (recentered cumulative score ≥ 975).

Figure B9: Correcting Manipulation versus Overcorrection



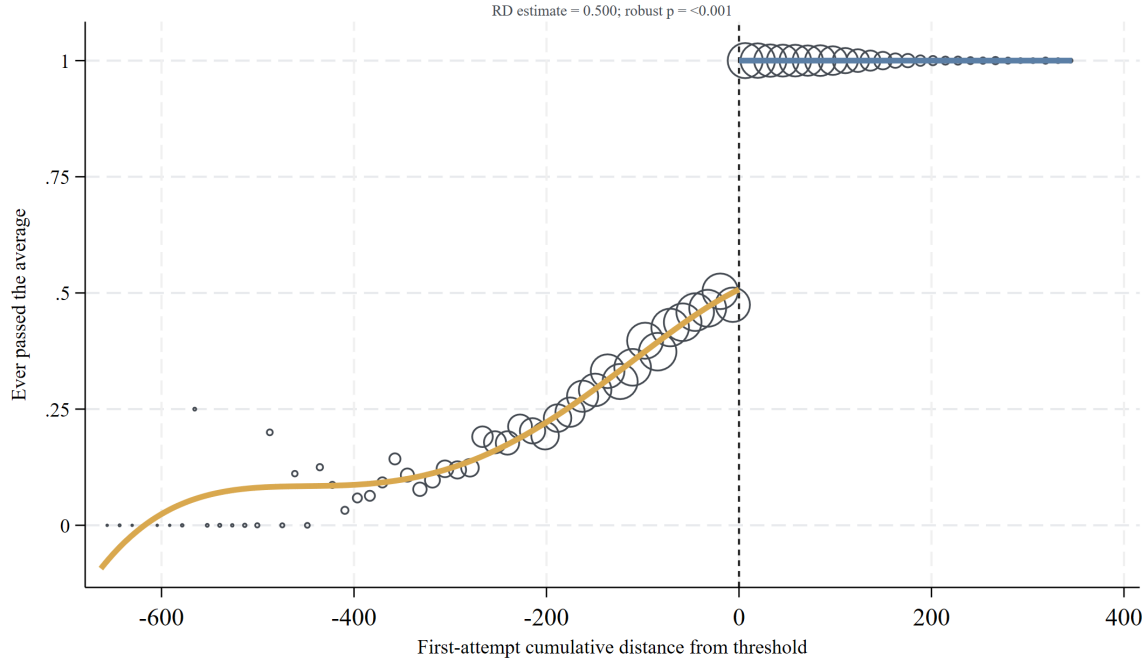
Notes: Panel (a) of Figure B7, enlarged around the passing threshold. Excluding candidates with a minimum subject score of exactly 60 removes the shaded purple region—the manipulated records bumped to the 60 floor, piled just above the cutoff—but also the shaded gold region, the genuine records that legitimately scored a minimum of 60.

Figure B10: Correcting Two Forms of Manipulation versus Overcorrection



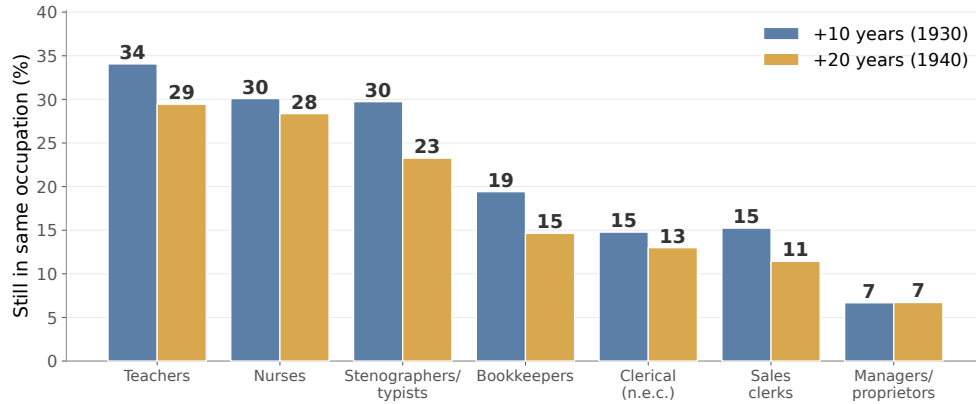
Notes: Panel (a) of Figure B7, enlarged around the passing threshold. Excluding candidates with a minimum subject score of exactly 60 removes three groups: candidates whose average passes only because a failing subject score is raised to 60 (purple), candidates who had already cleared the average and whose failing subject score is raised to 60 (teal), and candidates who genuinely scored a minimum of 60 (gold).

Figure B11: First Stage: Passing the Average on the First Attempt and Ever Passing

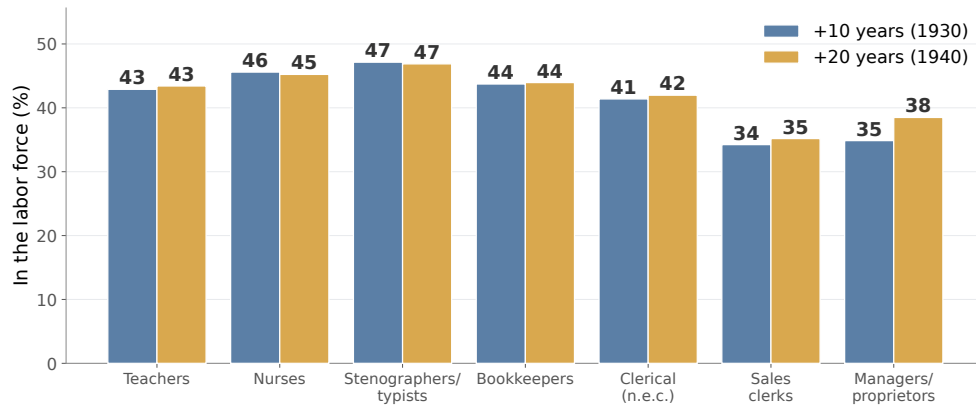


Notes: Regression-discontinuity plot of ever passing the average on the first-attempt cumulative distance from the 975 passing threshold, among women aged 17–25 whose first-attempt minimum subject score is not exactly 60. “Ever passed” equals one if the candidate’s cumulative cleared the 975 threshold in any of her first four sittings (the first attempt plus the next three). Hollow circles are binned means with area proportional to the number of applicants in the bin; the curves are fourth-order polynomial fits estimated separately on each side of the cutoff. Candidates scoring above the cumulative threshold on their first attempt passed, while roughly half of the marginal failures eventually cleared the cumulative threshold (first stage: 0.50; robust $p < 0.001$). Of all candidates who ever pass the average, about 60 percent do so on the first attempt.

Figure B12: Occupational retention: teaching versus other white-collar occupations



(a) Still in the same occupation



(b) In the labor force

Notes: 1920 cohort: white women aged 18–30 observed in the occupation in the 1920 census and linked to both the 1930 and 1940 censuses (national linked full-count panel). Panel (a) plots the share still recorded in the *same* occupation ten and twenty years later; panel (b) the share in the labor force in any occupation.

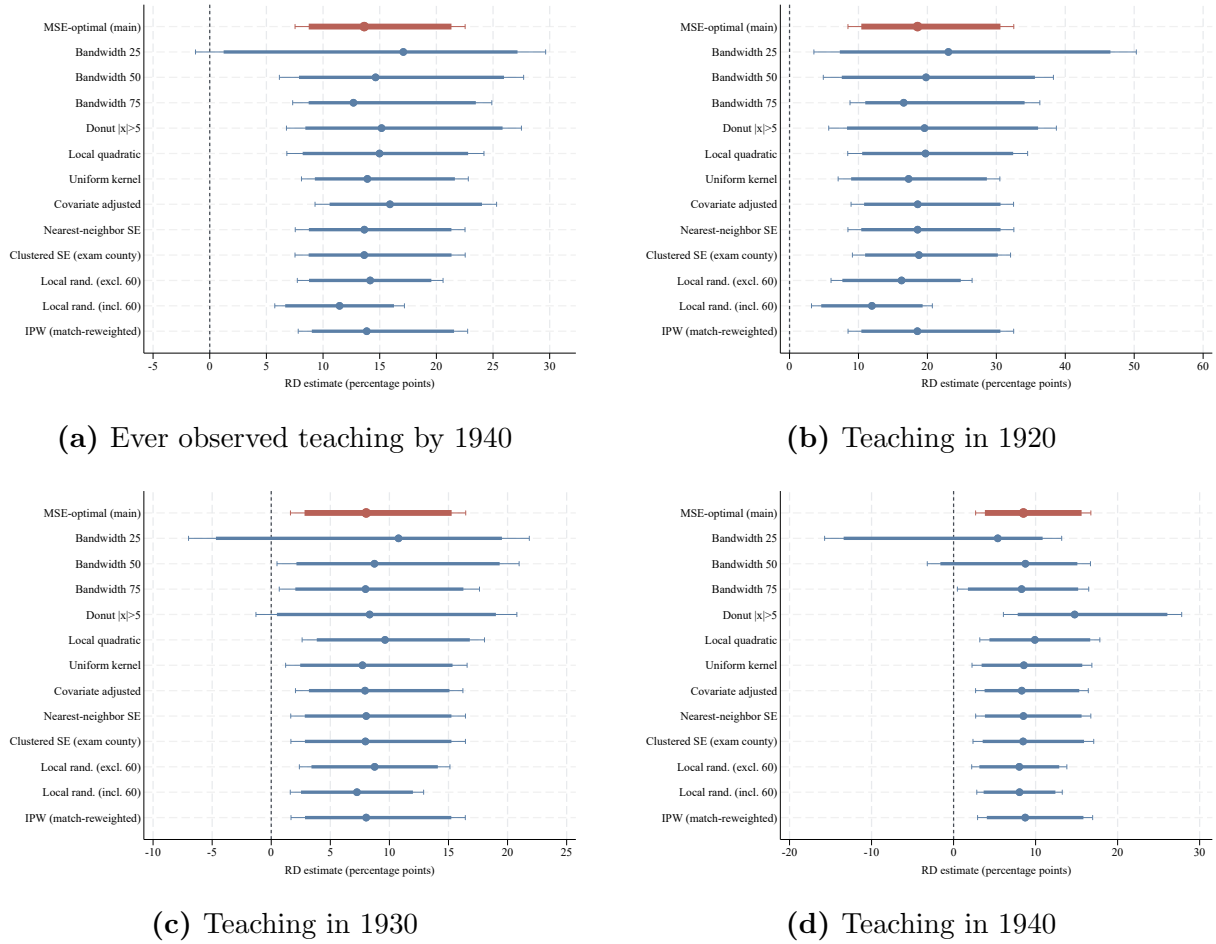
B.1 Robustness Across Specifications

Figures B13–B18 compare the main estimate with twelve alternative specifications, including standard errors clustered by exam county, local-randomization estimates with and without records whose minimum subject score is exactly 60, and an estimate that reweights the linked sample to its baseline composition. Every estimate keeps its sign across these specifications.²⁵ Magnitudes are particularly stable for the 1930 and 1940 teaching and labor-force outcomes; the short-run outcomes and income are somewhat more sensitive, while retaining the same

²⁵With the exception of the local randomization estimate for new education.

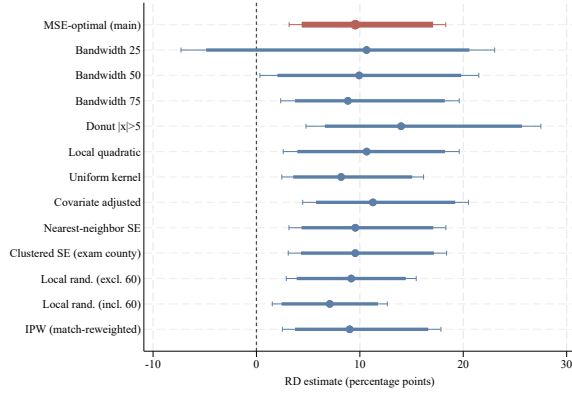
qualitative pattern.

Figure B13: Robustness: Teaching

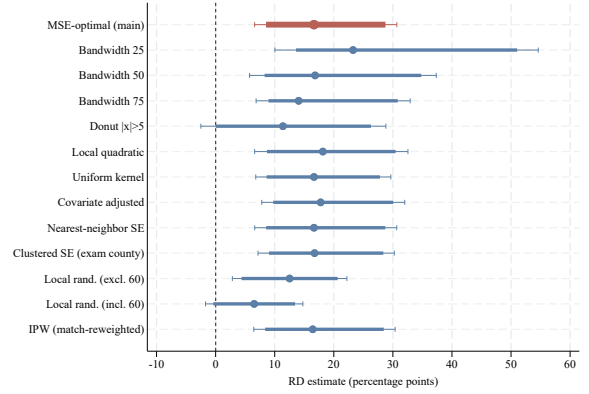


Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

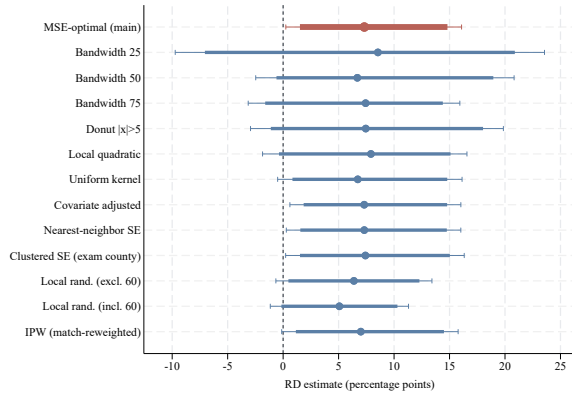
Figure B14: Robustness: Labor Force Participation



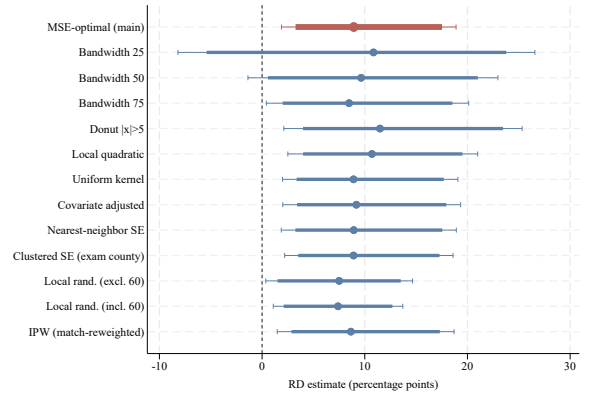
(a) Ever observed in the labor force by 1940



(b) Labor force participation in 1920



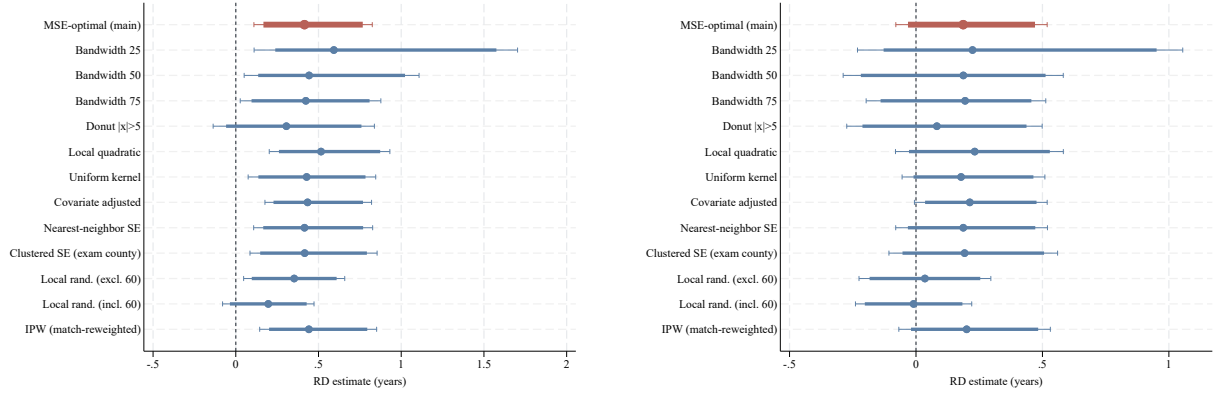
(c) Labor force participation in 1930



(d) Labor force participation in 1940

Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B15: Robustness: Education

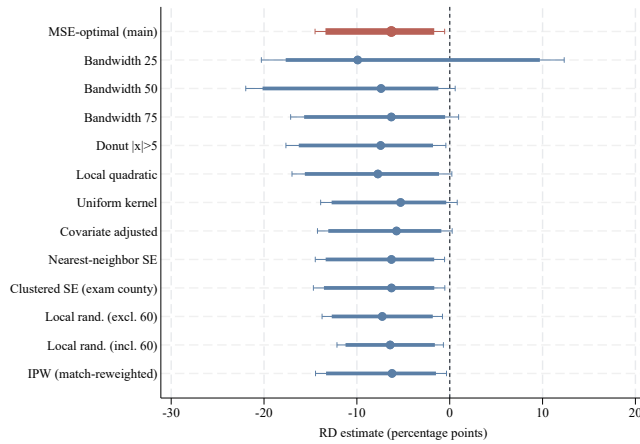
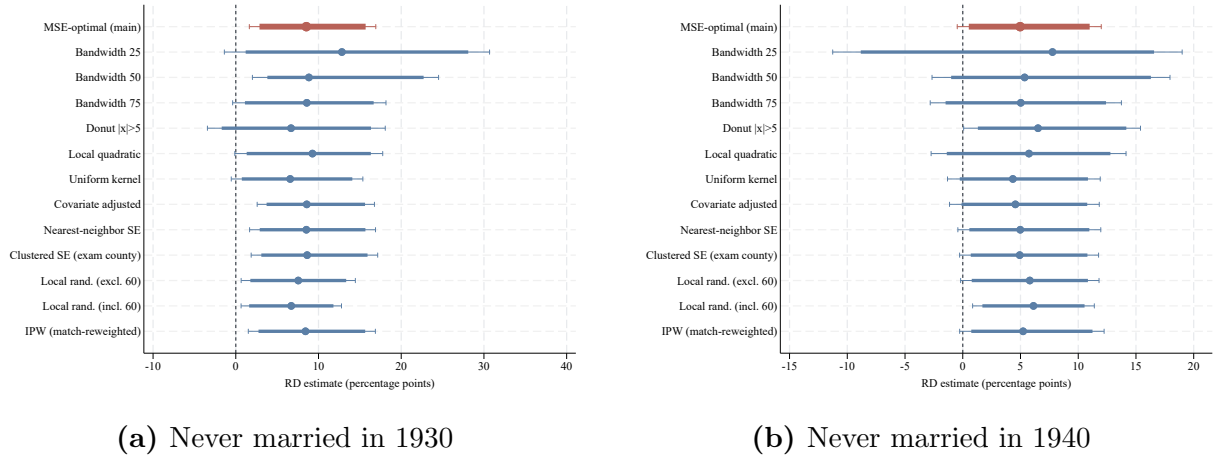


(a) Years of education in 1940

(b) New education since the exam

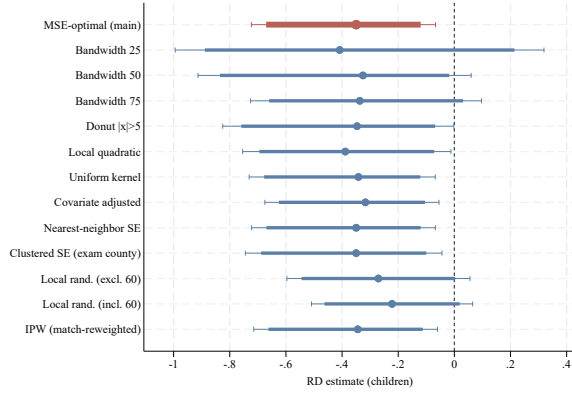
Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B16: Robustness: Marriage

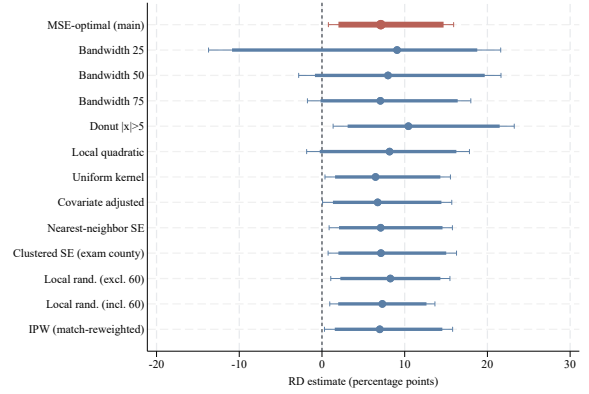


Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

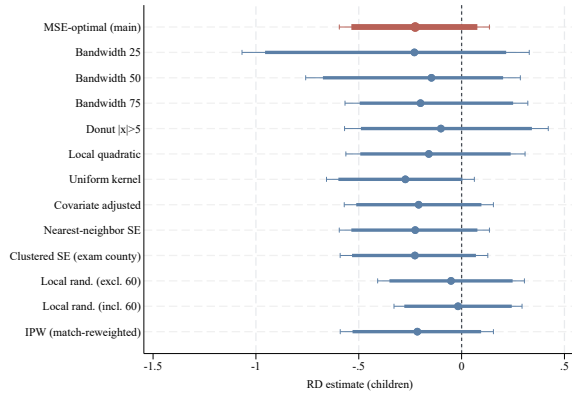
Figure B17: Robustness: Fertility



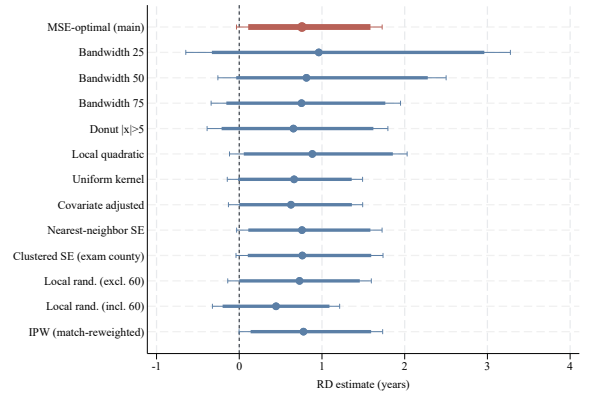
(a) Estimated fertility



(b) No children by 1940



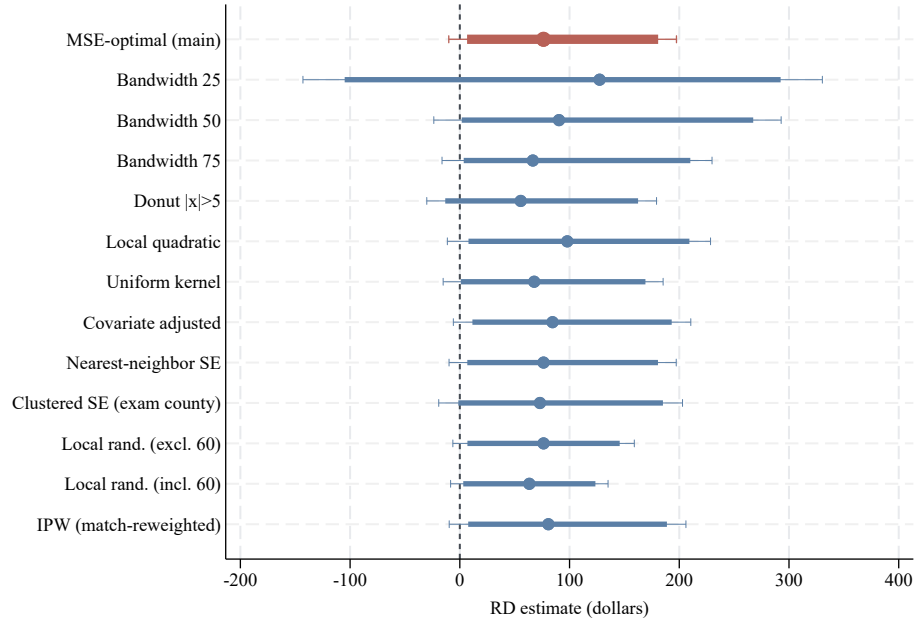
(c) Fertility conditional on positive



(d) Age at first birth

Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B18: Robustness: Own Wage Income in 1940

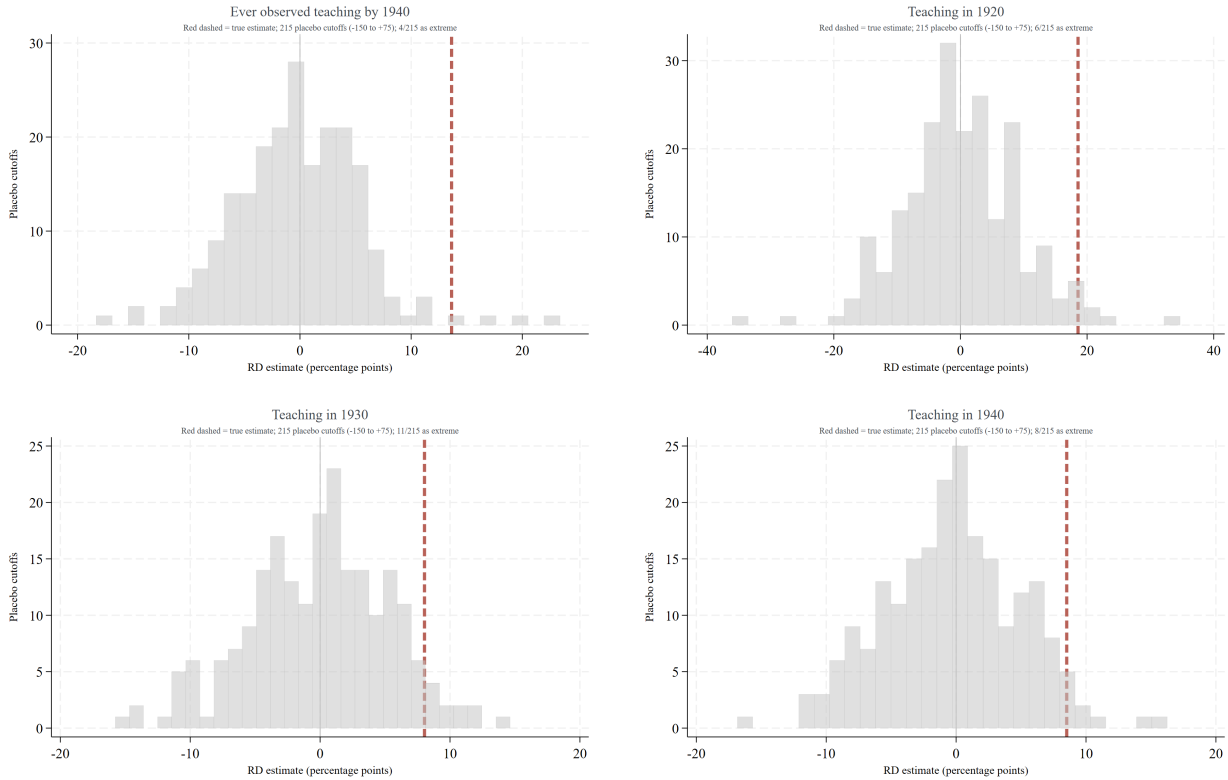


Notes: The red marker is the main estimate, using the MSE-optimal bandwidth. Blue markers are twelve alternatives: fixed bandwidths of 25, 50, and 75 points; a donut that drops observations within five points of the cutoff; a local-quadratic fit; a uniform kernel; controls for age at exam, education at exam, and birth year; nearest-neighbor standard errors; standard errors clustered by exam county; local-randomization estimates within 13 points of the cutoff (one average point), with and without records whose minimum subject score is exactly 60; and an estimate that reweights the linked sample to the baseline composition by the inverse of the census-match propensity. Markers show conventional point estimates. Thick bars are 90 percent and thin capped lines 95 percent confidence intervals, bias-corrected for the local-polynomial specifications and conventional for the local-randomization estimates. Binary outcomes are in percentage points and continuous outcomes in their natural units.

C Placebo Cutoff Distributions

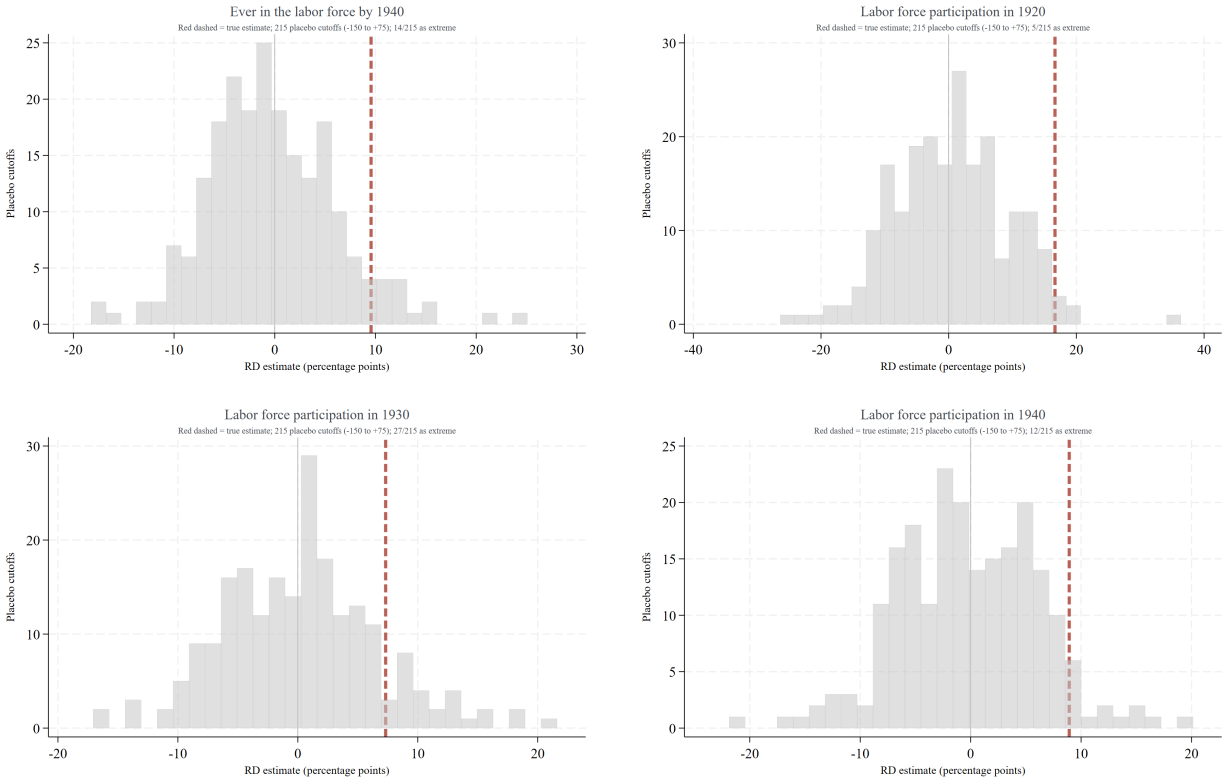
I re-estimate the RD across a range of false cutoffs and plot the true estimate against the distribution of placebo estimates. For each outcome I place a false cutoff at every integer score from 150 points below to 75 points above the true passing threshold, excluding a narrow window within 5 points of the true cutoff, and estimate the RD. Each placebo estimate uses only observations on the side of the true threshold to which the false cutoff is shifted, so the true discontinuity does not enter the placebo estimates (Imbens and Lemieux, 2008). This produces 215 placebo estimates per outcome.

Figure B19: Placebo-cutoff distributions: Teaching



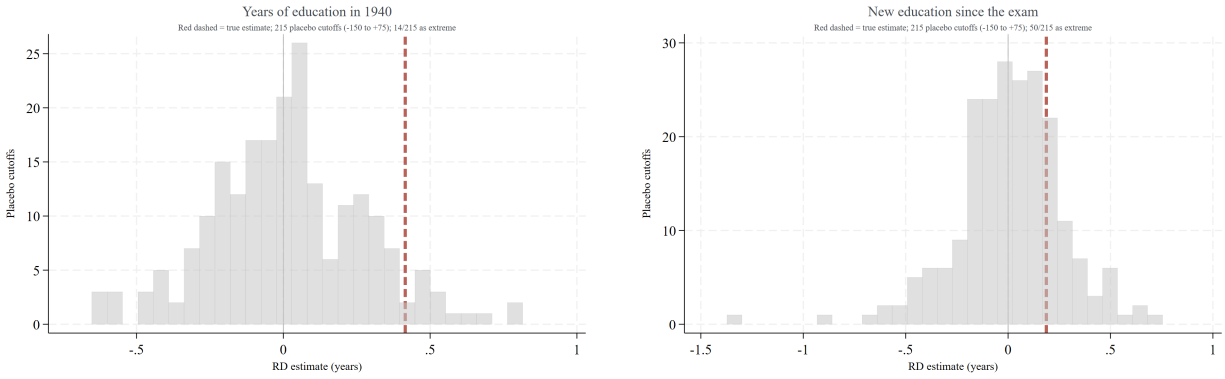
Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B20: Placebo-cutoff distributions: Labor Force Participation



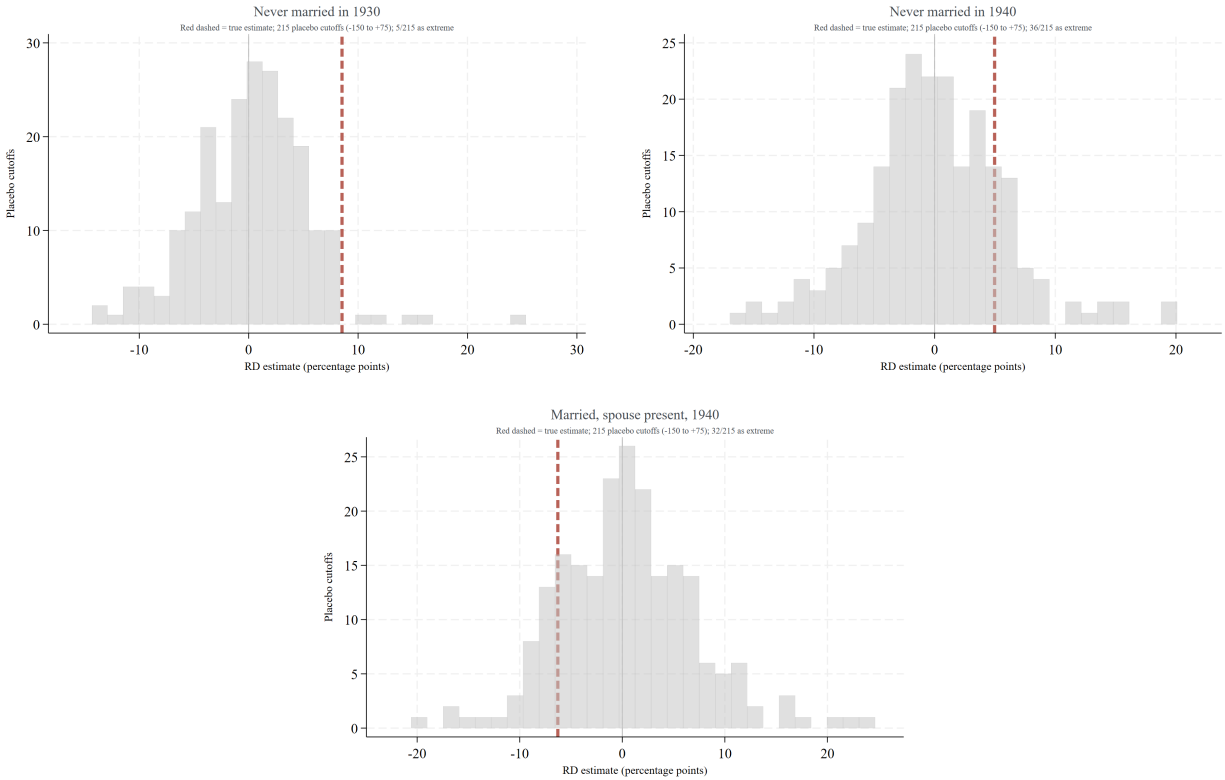
Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B21: Placebo-cutoff distributions: Education



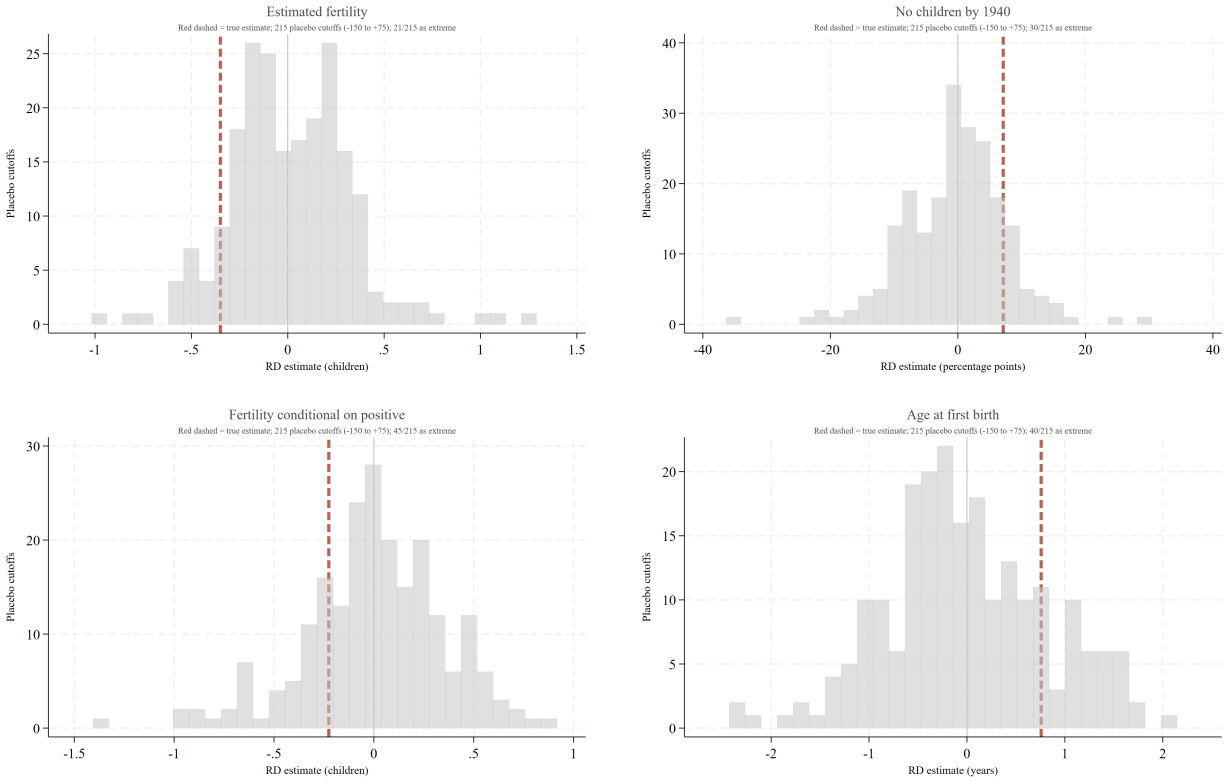
Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B22: Placebo-cutoff distributions: Marriage



Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B23: Placebo-cutoff distributions: Fertility



Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.

Figure B24: Placebo-cutoff distributions: Own Wage Income in 1940



Notes: Grey bars are the distribution of RD point estimates at 215 false cutoffs, placed at every integer score from 150 points below to 75 points above the true threshold (excluding a window within 5 points of it). The red dashed line is the true estimate at the certification threshold. Each false cutoff uses only observations on the side of the true threshold to which it is shifted, so the true discontinuity does not contaminate the placebo. Estimates use the MSE-optimal bandwidth (HC3 standard errors). Binary outcomes are in percentage points and continuous outcomes in their natural units.